

Reviewed Preprint

v1 • July 7, 2025

Not revised

Reviewed Preprint

v2 • February 12, 2026

Revised by authors

✉ For correspondence:

yangwen.xu@unitn.it

roberto.bottini@unitn.it

Competing interests: No

competing interests declared

Funding: See page 23

Reviewing editor: Xing Tian, New York University Shanghai, China

© 2025, Xu et al. This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Neural Representation of Time across Complementary Reference Frames

Yangwen Xu^{1,2}✉, Nicola Sartorato^{1,3,4}, Léo Dutriaux^{1,5}, Roberto Bottini¹✉

¹Center for Mind/Brain Sciences, University of Trento, Trento, Italy • ²Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany • ³Institute of Neurobiology, University of Tübingen, Tübingen, Germany • ⁴Werner-Reichardt Centre for Integrative Neuroscience, Tübingen, Germany • ⁵Laboratoire d'Études des Mécanismes Cognitionnels, Université Lumière Lyon 2, Lyon, France

eLife Assessment

This study presents a **valuable** finding on the neural representation of time from two distinct egocentric and allocentric reference frames. The evidence is **solid** and largely supports the hypothesis, with one caveat that the task differences could impact the observed effects. The work will be of interest to cognitive neuroscientists working on the perception and memory of time.

<https://doi.org/10.7554/eLife.107273.2.sa3>

Abstract

Humans conceptualize time in terms of space, allowing flexible time construals from various perspectives. We can travel internally through a timeline to remember the past and imagine the future (i.e., mental time travel) or watch from an external standpoint to have a panoramic view of history (i.e., mental time watching). However, the neural mechanisms that support these flexible temporal construals remain unclear. To investigate this, we asked participants to learn a fictional religious ritual of 15 events. During fMRI scanning, they were guided to consider the event series from either an internal or external perspective in different tasks. Behavioral results confirmed the success of our manipulation, showing the expected symbolic distance effect in the internal-perspective task and the reverse effect in the external-perspective task. We found that the activation level in the posterior parietal cortex correlated positively with sequential distance in the external-perspective task but negatively in the internal-perspective task. In contrast, the activation level in the anterior hippocampus positively correlated with sequential distance regardless of the observer's perspectives. These results suggest that the hippocampus stores the memory of the event sequences allocentrically in a perspective-agnostic manner. Conversely, the posterior parietal cortex retrieves event sequences egocentrically from the optimal perspective for the current task context. Such complementary allocentric and egocentric representations support both the stability of memory storage and the flexibility of time construals.

Introduction

How the brain represents time remains mysterious. In prospective timing, a starting point is defined beforehand, and the neural activity tracks the elapsed time like a stopwatch. This neural stopwatch is manifested in various forms, including ramping activity, sequential activity, and neural trajectories. (e.g., see reviews by Buonomano & Laje, 2010 [DOI](#); Wittmann, 2013 [DOI](#); Eichenbaum, 2014 [DOI](#); Tsao et al., 2022 [DOI](#)). However, a stopwatch can only track duration in real time. How can we escape from the present, being able to remember the past or imagine the future?

One solution, which might be unique to humans, is to conceptualize time in terms of space (i.e., the spatial construals of time hypothesis or the conceptual metaphor theory, e.g., Traugott, 1978 [DOI](#); Lakoff & Johnson 1980 [DOI](#); see recent reviews by Núñez & Cooperider, 2013 [DOI](#); Sinha & Gärdenfors,

2014 [\[1\]](#)). This is achieved by segmenting time into events—the basic temporal entities the observer conceives to have a beginning and an end (Zacks & Tversky, 2001 [\[2\]](#))—and ordering these temporal entities in space so that events occurring at different times can be easily maintained in the working memory (Abrahamse et al., 2014 [\[3\]](#); Figure 1A [\[4\]](#)). This reconstructive process consists of two core time concepts: duration and sequence.

Unlike prospective timing tracking the continuous passage of time, durations in time construals are event-based (Tsao et al., 2022 [\[5\]](#)): the intervals' boundaries are constituted by events (Sinha & Gärdenfors, 2014 [\[6\]](#)), and the duration of events reflect their span (Núñez & Cooperrider, 2013 [\[7\]](#); Figure 1A [\[4\]](#)). Neurocognitive evidence suggests that the neural representation of duration engages distinct brain systems. The motor system—particularly the supplementary motor area—has been associated with prospective timing (e.g., Protopapa et al., 2019 [\[8\]](#); Nani et al., 2019 [\[9\]](#); De Kock et al., 2021 [\[10\]](#); Robbe, 2023 [\[11\]](#)), whereas the hippocampus is considered to support the representation of duration embedded within an event sequence (e.g., Barnett et al., 2014 [\[12\]](#); Thavabalasingam et al., 2018 [\[13\]](#); see also the comprehensive review by Lee et al., 2020 [\[14\]](#)).

As for event sequence, one century ago, the philosopher John McTaggart proposed the distinction between two core time construals: the A-series and the B-series (McTaggart, 1908 [\[15\]](#)). The A-series assumes a deictic center of time—the observer's subjective now—as the reference, and orders events as being in the past or the future according to this deictic center. The B-series concerns the order of events in a sequence regardless of the observer's subjective now. The distinction between A- and B-series has been widely echoed in various accounts of the temporal frames of reference under different names in the cognitive linguistic analyses (see reviews by Bender & Beller, 2014 [\[16\]](#)).

In general terms, these two core time construals can be understood through two complementary perspectives: an internal and an external one (Núñez & Cooperrider, 2013 [\[7\]](#); Tversky & Jamalian, 2021 [\[17\]](#); Figure 1A [\[4\]](#)). The internal perspective on time series is akin to the “route” perspective in the spatial domain (Siegel & White, 1975 [\[18\]](#)). It aligns with the cognitive process called “mental time travel” (Tulving, 1984 [\[19\]](#), 2002 [\[20\]](#); Suddendorf et al., 2009 [\[21\]](#)). The traveler can project themselves into any event in the time series and redefine past and future according to their self-location—their subjective now. In this sense, the brain is no longer a stopwatch but a time machine, taking the traveler back and forth in time (Buonomano, 2017 [\[22\]](#)). The A-series is typically constructed from this internal viewpoint. By contrast, the external perspective on time series is akin to the “survey” perspective in the spatial domain (Siegel & White, 1975 [\[18\]](#)). It relates to the cognitive process called “mental time watching” (Stocker, 2012 [\[23\]](#), 2014 [\[24\]](#)). The watcher, who is outside of the time series, can have a panoramic view of multiple events at different times and localize them relative to one another or to external temporal landmarks (e.g., sunrise and sunset in a day or historical events in the long term). In this sense, the brain is more like a dimensional ascension device, taking the watcher out of the one-dimensional timeline to an external viewpoint in higher-dimensional space. The B-series is generally constructed from this external viewpoint.

Recent studies have already begun to investigate the neural representation of the memorized event sequence (e.g., Deuker et al., 2016 [\[25\]](#); Thavabalasingam et al., 2018 [\[13\]](#); Bellmund et al., 2019 [\[26\]](#), 2022 [\[27\]](#); see reviews by Cohn-Sheehy & Ranganath, 2017 [\[28\]](#); Bellmund et al., 2020 [\[29\]](#)). Yet, the neural mechanisms that enable the brain to construct distinct construals of an event sequence remain largely unknown. Valuable insights may be drawn from research in the spatial domain, which differentiates the neural representation in allocentric and egocentric reference frames. According to an influential neurocomputational model (Byrne et al., 2007 [\[30\]](#); Bicanski & Burgess, 2018 [\[31\]](#); Bicanski & Burgess, 2020 [\[32\]](#)), allocentric and egocentric spatial representations are dissociable in the brain—they are respectively implemented in the medial temporal lobe (MTL)—including the hippocampus—and the parietal cortex. Various egocentric representations in the parietal cortex derived from different viewpoints can be transformed and integrated into a unified allocentric representation and stored in the MTL (i.e., bottom-up process). Conversely, the allocentric representation in the MTL can serve as a template for reconstructing diverse egocentric representations across different viewpoints in the parietal cortex (i.e., top-down process).

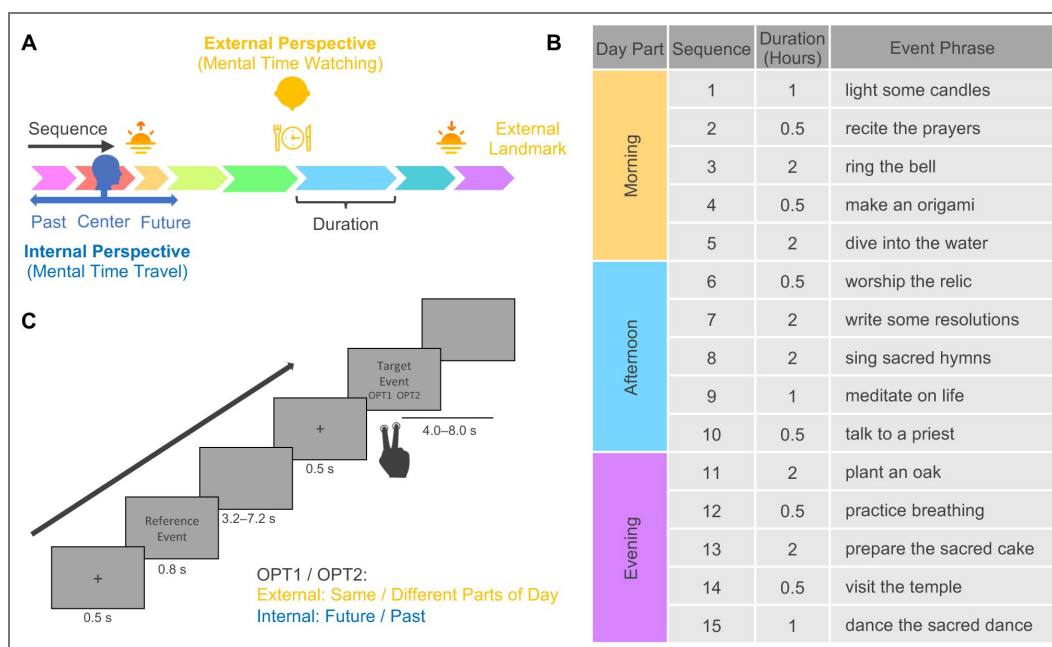


Figure 1. Spatial construal of time and experimental design.

(A) The schematic diagram of spatial construal of time. It illustrates two core time concepts (sequence and duration) and two major perspectives on event series (mental time travel and watching). (B) Stimulus: a fictional religious ritual of 15 events following a specific sequence, enduring particular durations, and happening on predetermined parts of the day. To minimize potential confounds between the semantic content of the event phrases and the temporal structure of the events, we randomly assigned the phrases to the events, creating two versions for participants with even and odd ID numbers. Only one version is illustrated here. Both versions can be seen in [Figure 1—figure supplement 1](#) and [Figure 1—source data 1](#). (C) Task paradigm. In the external-perspective task, participants judged whether the target events happened in the same part of the day as the reference event. In the internal-perspective task, participants imagined themselves doing the reference events and judged whether the target event happened in the past or will happen in the future.

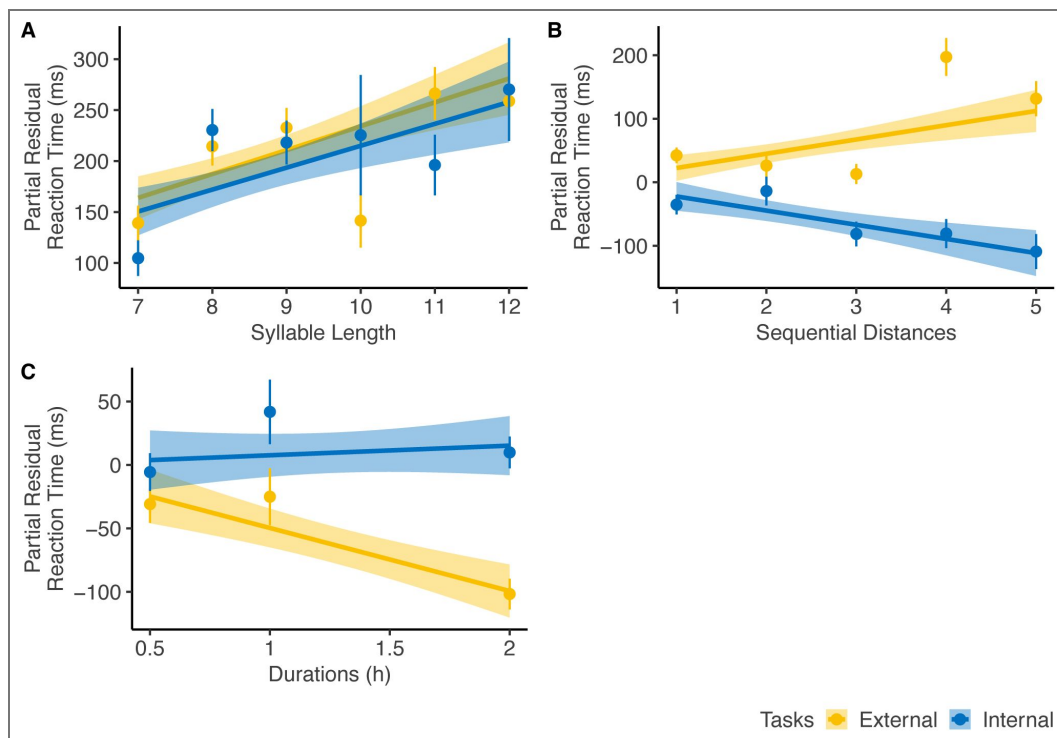


Figure 2. Reaction time analysis.

For diagnostic purposes, we plotted the partial residuals of each significant predictor significantly influencing the reaction time (RT) of the corrected trials. The partial residual includes the effect of each variable, its interaction with *Task Type*, and the residuals from the full linear regression model. A. RT increased with *Syllable Length*, showing a similar trend across both tasks. B. *Sequential Distance* affected RT in opposite directions depending on the perspectives. C. *Event Duration* influenced RT only in the external-perspective task, with no effect in the internal-perspective task.

In line with the spatial construals of time hypothesis, several authors have recently suggested that such mutually engaged egocentric and allocentric reference frames (in the parietal cortex and the medial temporal lobe, respectively) proposed in the spatial domain might also apply to the temporal one (e.g., Gauthier & van Wassenhove, 2016a [b](#); Gauthier et al., 2019 [c](#), 2020 [c](#); Bottini & Doeller, 2020 [c](#)). If this hypothesis holds, it could explain how the brain flexibly generates diverse construals of the same event sequence. Specifically, the hippocampus may encode a consistent representation of an event sequence that is independent of whether an individual adopts an internal or external perspective, reflecting an allocentric representation of time. In contrast, parietal cortical representations are expected to vary flexibly with the adopted perspective that is shaped by task demands, reflecting an egocentric representation of time.

This functional magnetic resonance imaging (fMRI) study aimed to directly test this hypothesis by systematically investigating the neural mechanisms underlying the time construals of event sequence and duration. The event series was a fictional religious ritual of 15 events that participants learned the day before scanning (Figure 1B [c](#)). Such a ritual had all the core temporal elements: the constituent events followed a specific sequence, endured particular durations, and happened on predetermined parts of the day (i.e., the external temporal landmarks). Participants learned these core temporal elements by reading the description and imagining going through the events one after another. The post-learning test suggests that all participants learned the temporal structure of the ritual before scanning (see Methods for details of the learning and testing procedure).

During the fMRI scanning, participants performed two tasks guiding them to consider the event series from internal and external perspectives (i.e., mental time travel vs. mental time watching; Figure 1C [c](#)). In each trial, participants saw two sequential event phrases. The first was the reference event, and the second was the target event. In the external-perspective task, participants localized events relative to external temporal boundaries, judging whether the target event happened in the same or different part of the day as the reference event. In the internal-perspective task, participants were instructed to project themselves into the reference event and localize the target event relative to temporal point, judging whether the target event happened in the future or the past of the reference event (see Methods for details of the scanning procedure).

Results

Time was processed differently from internal and external perspectives

Participants had significantly greater accuracy in the external-perspective task than the internal-perspective task (external-perspective task: $M = 93.5\%$, $SD = 4.7\%$; internal-perspective task: $M = 89.5\%$, $SD = 8.1\%$; paired $t(31) = 3.33$, $p = 0.002$). The reaction time (RT) of the corrected trials in the external-perspective task was also significantly shorter than the internal-perspective task (external-perspective task: $M = 1475$ ms, $SD = 529$ ms; internal-perspective task: $M = 1578$ ms, $SD = 587$ ms; fixed effect of *Task Type* in a random-intercept-and-slope linear mixed model with *Participant* as the random-effect grouping factor: $F(1, 31) = 27.44$, $p < 0.001$).

To further explore the factors affecting the RT of the correct trials, we built a random-intercept linear mixed model with *Participant* as the random effects grouping factor. Fixed effects variables included *Sequential Distance* (i.e., number of events between the reference and the target events), *Duration* (i.e., duration of the target events), *Syllable Length* (i.e., number of syllables of the phrase of the target events), *Task Type* (i.e., external-vs. internal-perspective tasks), and the interaction between *Task Type* and all the other variables (Table 1 [c](#)). As a sanity check, we found a significant main effect of *Syllable Length* ($F(1, 6918) = 35.59$, $p < 0.001$) since it was expected that participants spent longer time reading longer phrases. Intriguingly, we also found significant interaction effects between *Task Type* and *Sequential Distance* ($F(1, 6918) = 28.22$, $p < 0.001$) and *Task Type* and *Duration* ($F(1, 6918) = 12.81$, $p < 0.001$).

Fixed Effects ²	df	F	p
Sequential Distance	1, 6918	0.0003	0.987
Duration	1, 6918	6.914	0.009
Syllable Length	1, 6918	35.585	< 0.001
Task Type	1, 6918	4.357	0.037
Task Type × Sequential Distance	1, 6918	28.224	< 0.001
Task Type × Duration	1, 6918	12.809	< 0.001
Task Type × Syllable Length	1, 6918	0.065	0.799

Table 1. The reaction time of the corrected trials indicates that time is differently processed under internal and external perspectives¹.

¹Linear Mixed Model Formula: $RT \sim 1 + \text{Task Type} * (\text{Sequential Distance} + \text{Duration} + \text{Syllable Length}) + (1 | \text{Participant})$ ²The significant effects were highlighted. We did not highlight the significant main effects if the corresponding interaction effects were also significant.

Sequential Distance was correlated positively with RT in the external-perspective task ($z = 3.80$, $p < 0.001$) but negatively in the internal-perspective task ($z = -3.71$, $p < 0.001$). The negative correlation between RT and egocentric distance has been consistently observed in previous studies in which participants engaged in temporal self-projection: participants make more effort to differentiate past and future for the events close to their own temporal position (e.g., Arzy et al., 2008 [b](#); Arzy et al., 2009a [b](#); Gauthier & Wassenhove, 2016a [b](#); Gauthier et al., 2019 [b](#)). This pattern can broadly be attributed to the symbolic distance (SD) effect (Moyer & Landauer, 1967 [b](#); Moyer & Bayer, 1976 [b](#); Shepard & Judd, 1979), which refers to the fact that “time needed to compare two symbols varies inversely with the distance between their referents on the judged dimension” (Moyer & Bayer, 1976 [b](#), p. 229). The positive correlation between RT and sequential distance from an external perspective was predicted by Gauthier & van Wassenhove (2016a [b](#)). This prediction was inspired by classic studies on mental scanning using visual imagery (e.g., Shepard & Metzler, 1971; Kosslyn et al., 1978 [b](#)): From an external perspective, people might compare two referents by mentally traversing the intermediate states between them, resulting in longer times to scan longer sequential distances.

As for *Duration*, it had a significantly negative trend with RT in the external-perspective task ($z = -4.44$, $p < 0.001$) but not in the internal-perspective task ($z = 0.66$, $p = 0.51$). A possible explanation is that events with longer durations were more salient and thus easier to be compared with external landmarks.

To ensure that the behavior outputs in neither internal- nor external-perspective tasks confounded the effects above, we built another linear mixed model incorporating four additional variables: Same/Different parts of the day, Future/Past, and their respective interactions with *Task Type* (Supplementary File 1). The interaction effects between *Task Type* and both *Sequential Distance* and *Duration* remained significant.

Furthermore, we found an additional interaction effect between *Task Type* and *Same/Different* parts of the day ($F(1, 6914) = 70.49$, $p < 0.001$). This effect was in line with the effect of *Sequential Distance*, with two events in the same and different parts of the day corresponding to the short and long sequential distances, respectively. This effect paralleled that of *Sequential Distance*, as events occurring within the same or different parts of the day corresponded to shorter and longer sequential distances, respectively. This pattern can be interpreted in terms of a categorical effect: sequential distances within the same day part were perceived as shorter (i.e., a chunking effect), whereas distances spanning different day parts were perceived as longer (i.e., a boundary effect).

The behavioral results suggest that our attempt to induce different perspectives on the event series was successful. The two tasks induced RT as distinct functions of sequential distance, in line with the predicted SD effect for the internal perspective and reverse-SD effect for the external perspective (Gauthier & van Wassenhove, 2016a [b](#)).

Internal-compared to external-perspective task activated different brain networks

We first directly contrasted the activity level between external- and internal-perspective tasks in the time window of the target events (Figure 3A [b](#); Table 2 [b](#); see Figure 3—figure supplement 1A [b](#) for a surface view; voxel level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). Compared with the external-perspective task, the internal-perspective task specifically activated the regions in the default network (DN) in the right hemisphere. They were the precuneus (PreC), the retrosplenial cortex (RSC), the superior frontal gyrus (SFG), and the angular gyrus (AG). This finding aligned with evidence indicating that the DN plays a crucial role in self-projection (see the review by Buckner & Carroll, 2007 [b](#)). In Figure 3—figure supplement 1 [b](#), we also compared the significant clusters in the internal-perspective task with the two subnetworks of the DN: DN-A and DN-B (e.g., Braga & Buckner, 2017 [b](#); Reznik et al., 2013). The internal-perspective mostly engages DN-A rather than DN-B. This observation is consistent with existing evidence suggesting that DN-A is more closely associated with episodic memory, whereas DN-B is primarily involved in social processing (e.g., Lin et al., 2018 [b](#); DiNicola et al., 2020 [b](#)).

Figure 3. Neural correlates of specific perspectives and syllable length.

(A) Univariate contrast between external-perspective and internal-perspective tasks (voxel-level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). All the significant areas were in the right hemisphere. PreC: precuneus; RSC: the retrosplenial cortex; SFG: the superior frontal gyrus; AG: angular gyrus; SMA: supplementary motor area; SMG: supramarginal gyrus. (B) Parametric modulation of syllable length as a sanity check (voxel-level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). The activation level in the anterior part of the left superior temporal gyrus and the visual cortex positively correlated with syllable length. R = Right Hemisphere; L = Left Hemisphere.

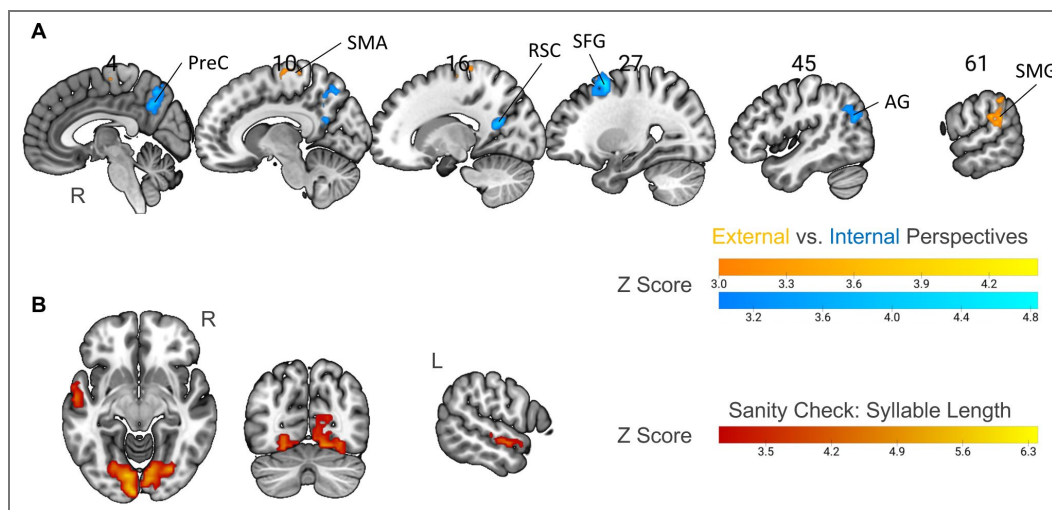


Table 2. Univariate contrast between internal- and external-perspective tasks ($p < 0.001$, cluster-level FWE corrected $p < 0.05$ across the whole cortex).

¹The average Montreal Neurological Institute coordinates of all the significant voxels of each cluster. The precuneus and the retrosplenial cortex were connected as one cluster under the threshold $p < 0.001$ ($z > 3.09$). In this case, we increased the threshold to the point when the precuneus and the retrosplenial cortex were separate ($z > 3.3$) and calculated the average coordinates of each cluster. ²The voxel size is $3 \times 3 \times 3 \text{ mm}^3$.

Anatomical Label	Center MNI Coordinate ¹			Cluster Size
	X	Y	Z	Number of Voxels ²
Internal > External-Perspective Task				
Supplementary Motor Area (R)	10	-22	69	34
Supramarginal Gyrus (R)	61	-36	26	98
Internal > External-Perspective Task				
Precuneus (R)	4	-61	45	79
Retrosplenial Cortex (R)	16	-56	19	45
Superior Frontal Gyrus (R)	27	12	56	95
Angular Gyrus (R)	45	-66	30	78

Compared with the internal-perspective task, the external-perspective task specifically activated the supplementary motor area (SMA) and the supramarginal gyrus (SMG) in the right hemisphere (Figure 3A; Table 2; voxel level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). This finding is open to interpretation. The area found in the right SMG was centered at the junction region between the posterior part of the SMG and the posterior part of the superior temporal gyrus. Previous evidence has shown that this temporoparietal junction area relates to the out-of-body experience (e.g., see reviews by Blanke et al., 2004; Blanke & Arzy, 2005) or plays a role in overcoming egocentric emotion bias towards others (e.g., Silani et al., 2013). Thus, the right SMG region found here may be important for the mental construction of an external perspective.

The right posterior parietal cortex implemented opposite representations of sequential distance across external and internal perspectives

Next, we used the parametric modulation method to detect neural correlates of the temporal information (i.e., *Sequential Distance* and *Duration*) across external- and internal-perspective tasks. We built a single general linear model (GLM) in which the target events were simultaneously modulated by their duration and their sequential distances to the reference events. The target events in external- and internal-perspective tasks were treated separately. To detect whether the temporal information was represented differently from different perspectives, we first examined the interaction effect between each temporal information and the task type. If the interaction effect was not significant, we also examined the main effect combining both tasks.

As a sanity check, we investigated whether the employed parametric modulation method could successfully reveal the effect of the word length in the visual cortex, given that our stimuli were visually presented. To do so, we used *Syllable Length* as the parameter to modulate the condition of both the reference and the target events in the above GLM. We found not only the visual cortex but also the left superior temporal gyrus in the language network, of which the activity level positively correlated with the number of syllables (Figure 3B; voxel level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). This result confirmed our prediction and validated our methodology.

To detect the regions in which the activation level was modulated by *Sequential Distance* (i.e., the number of events between the reference and the target events), we first searched regions with a significant interaction effect between *Task Type* (i.e., external-vs. internal-perspective tasks) and *Sequential Distance* (Figure 4A-C). The only significant region we could find across the whole cortex was localized in the border area between the angular gyrus and the superior division of the lateral occipital cortex in the right hemisphere (i.e., the boundary area between Brodmann area 7, 19, and 39; Figure 4A; voxel level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). More specifically, this region is mostly at the lateral wall of the posterior intraparietal sulcus (hIP5: 56.2%, hIP6: 9.5%, hIP4: 5.9%; Richter et al., 2019) and the posterior part of the angular gyrus (PGp: 21.4%; Caspers et al., 2006, 2008) (assignment based on maximum probability map; Eickhoff et al., 2005). The MNI coordinate of the center voxel is 38, -69, 35.

For convenience, we will refer to this region as the posterior parietal cortex (PPC) in the following text. Figure 4B further shows the beta estimates of *Sequential Distance* in the parametric modulation analysis in the right PPC. The activation level in this region correlated with sequential distance positively in the external-perspective task ($t(31) = 2.97$, $P = 0.006$) but negatively in the internal-perspective task ($t(31) = -4.19$, $P < 0.001$).

Here, changes in activity levels within the PPC were found to align with RT. Whether to control for RT's influence on fMRI activation represents a well-known paradox. On the one hand, RT reflects underlying cognitive processes and therefore should not be fully controlled for. On the other hand, RT can independently influence neural activity, as several brain networks vary with RT irrespective of the specific cognitive process involved—a domain-general effect. For instance, regions within the multiple-demand network are often positively correlated with RT and task difficulty across diverse cognitive domains (e.g., Fedorenko et al., 2013; Mumford et al., 2024). To evaluate the second possibility, we conducted an additional control analysis by including trial-

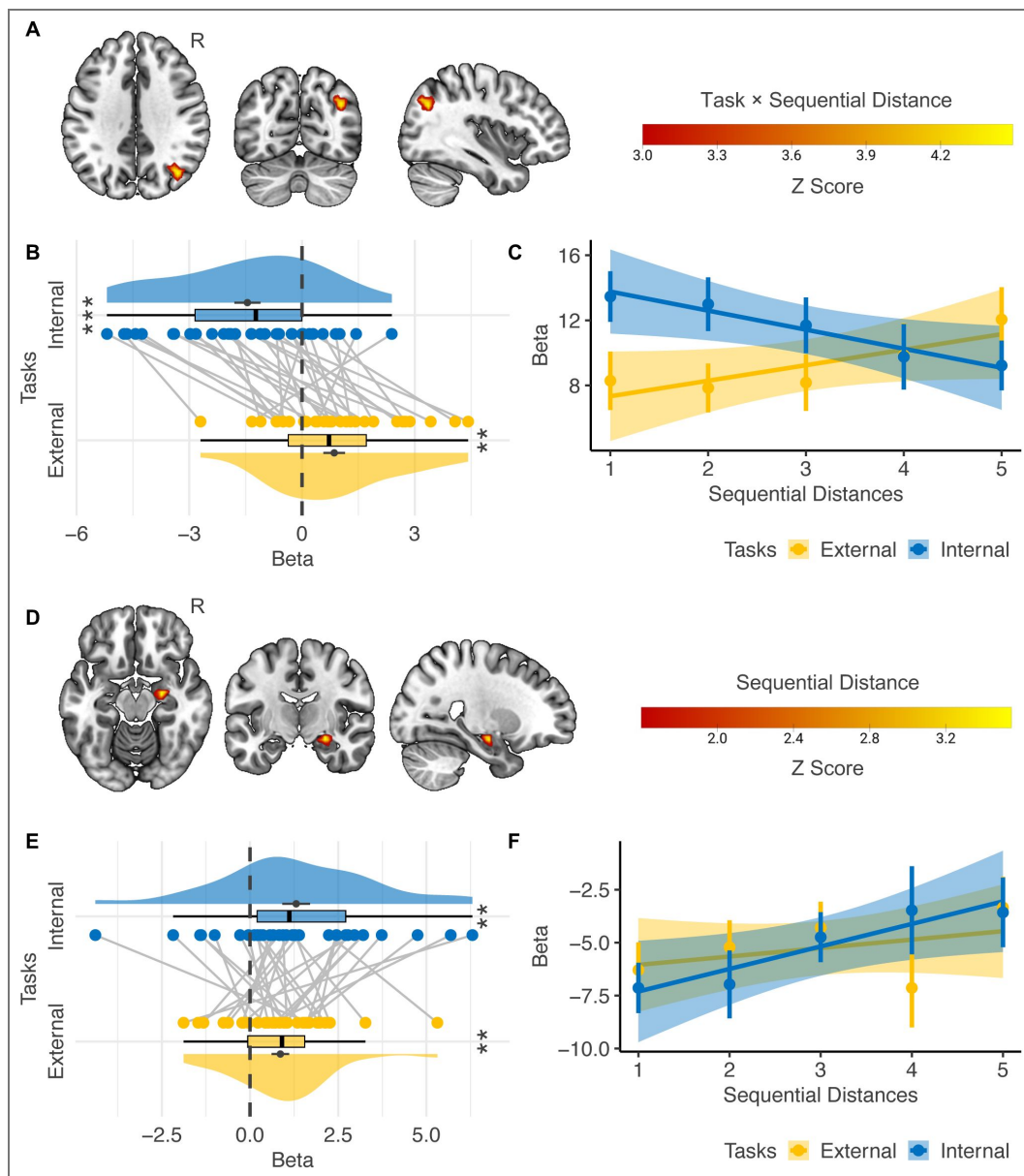


Figure 4. Neural correlates of event sequence.

A-C: Interaction effect between *Task Type* (i.e., external-vs. internal-perspective tasks) and *Sequential Distance*. (A) The only cortical region showing a significant interaction effect was localized in the right posterior parietal cortex (voxel-level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). (B) Regions of interest analysis shows that the activation level in the right posterior parietal cortex correlated with sequential distance positively in the external-perspective task and negatively in the internal-perspective task. (C) A further illustration of the relations between the activation level in the right posterior parietal cortex and sequential distance in the two tasks. The error bar indicates the standard error relative to the mean, and the shaded band around the linear regression line indicates 95% confidence interval. D-F: main effect of *Sequential Distance*. (D) The right hippocampal head shows a significant main effect of *Sequential Distance* within the mask of the bilateral hippocampus (voxel-level $p < 0.001$, cluster-level FWE corrected $p < 0.05$; voxel-level $p < 0.05$ for illustration purposes). (E) Regions of interest analysis shows that the correlation between the activation level in the right hippocampal head and the sequential distance was independent of perspectives. (F) A further illustration of the relations between the activation level in the right hippocampal head and sequential distance in the two tasks. The error bar indicates the standard error relative to the mean, and the shaded band around the linear regression line indicates 95% confidence interval. R = Right Hemisphere. **: $p < 0.01$; ***: $p < 0.001$.

by-trial RT as a parametric modulator in the first-level model (see Methods). Notably, the same PPC region remained the only area in the entire brain showing a significant interaction between Task Type and Sequential Distance (voxel-level $p < 0.001$, cluster-level FWE-corrected $p < 0.05$). This finding indicates that PPC activity cannot be fully attributed to RT. Furthermore, we do not interpret the effect as reflecting a domain-general RT influence, as regions within the multiple-demand system—typically sensitive to RT and task difficulty—did not exhibit significant activation in our data.

To provide a direct illustration of how the activation level in the right PPC varied according to the sequential distances in the two perspectives, we built another GLM with each sequential distance (i.e., 1 to 5) in each task as a separate condition. Figure 4C [confirms](#) that the activation level in the external-perspective task went up as the sequential distance increased, and the activation level in the internal-perspective task went down as the target event moved far away from the reference event where participants self-projected themselves.

These results suggest that the parietal cortex implements an egocentric representation of the event sequence, which varies with different perspectives.

The right hippocampal head implemented consistent representations of sequential distance across external and internal perspectives

We did not find any regions in the hippocampus showing a significant interaction effect between *Task Type* and *Sequential Distance*. Instead, we found a region in the right hippocampal head in which the activation level positively correlated with the sequential distance when we combined external- and internal-perspective tasks (Figure 4D [confirms](#); voxel level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). The MNI coordinate of the center voxel was 25, -11, -15. Figure 4E [further](#) shows that we did not find that such a positive correlation with the sequential distance significantly differed in the external- and internal-perspective tasks (paired $t(31) = -0.906$, $p = 0.372$).

We also illustrate the activation level in this hippocampal region of each sequential distance in external- and internal-perspective tasks, respectively (Figure 4F [confirms](#)): the activation level in the right hippocampus tended to go up as the sequential distance increased regardless of the tasks.

These results suggest that the hippocampus implements an allocentric representation of the event sequence independent of the perspectives, contrary to the egocentric representation in the posterior parietal cortex.

The right hippocampal body implemented the representation of event duration in the internal-perspective task

In the cortex, we did not detect any regions where the activation level showed a significant interaction effect between *Task Type* and *Duration* or a positive main effect of *Duration*. In the hippocampus, we also found no regions where the activation level showed a significant interaction effect between *Task Type* and *Duration*. Instead, we found a region in the right hippocampal body where the activation level showed a significantly positive correlation with *Duration* when combining two tasks (Figure 5A [confirms](#); voxel level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). The MNI coordinate of the center voxel is 39, -24, -12. However, Figure 5B [confirms](#) shows that directly comparing the beta estimates in the two tasks reveals a significant result (paired $t(31) = 3.07$, $p = 0.004$). The mean activation level in this area positively correlated with *Duration* only in the internal-perspective task ($t(31) = 5.54$, $p < 0.001$), not in the external-perspective task ($t(31) = 0.68$, $p = 0.502$). This later analysis is circular, as the ROI was defined as the voxels showing a significantly positive correlation with *Duration* combining two tasks in the first place. Nevertheless, it illustrated that this positive correlation in the voxel-level analysis was

driven only by the internal-perspective task. Due to the stringent threshold of multiple comparisons across voxels, this interaction effect between *Task Type* and *Duration* was not found in the voxel-level analysis.

Figure 5C [directly illustrates](#) how the activation level in this hippocampal region varied according to duration: the activation level in the right hippocampal body went up as the duration increased only in the internal-perspective task but not in the external-perspective task.

The difference in duration representation between the two tasks remains open to interpretation. One possible explanation is that the hippocampus is preferentially involved in memory for durations embedded within event sequences (see review by [Lee et al., 2020](#)). In the internal-perspective task, participants indeed localized events within the event sequence itself. By contrast, the external-perspective task encouraged participants to compare the event sequence with external temporal landmarks, which may have attenuated the hippocampal representation of duration.

We also implemented a post hoc analysis to fully illustrate the *Sequential Distance* and *Duration* effects in the internal-perspective task within the regions of interest of the hippocampal *Head* and the *Body* (Figure 5D [Figure 5D](#)). These two regions were defined as the regions where activation level significantly positively correlates with *Sequential Distance* and *Duration*, respectively. No significant evidence shows that the hippocampal head also represented *Duration* ($t(31) = 0.506$, $p = 0.616$), and no significant evidence shows that the hippocampal body also represented *Sequential Distance* ($t(31) = -0.616$, $p = 0.543$). Further confirmatory analyses are needed to verify such a double dissociation pattern between the neural representation of sequence and duration in the head and the body of the hippocampus.

Discussion

This study investigated the neural correlates of sequence and duration in task contexts where participants took internal or external perspectives on event series, thereby testing the complementary allocentric and egocentric reference frames in the temporal domain. We found that both the right PPC and the right hippocampal head represented the sequential distance between the reference and the target events. However, the representation in the right PPC varied with the perspective taken; its activity level correlated with sequential distance positively in the external perspective task and negatively in the internal perspective task. In contrast, the activation level in the right hippocampal head positively correlated with sequential distance regardless of the perspective taken. Moreover, we found that the activation level in the right hippocampal body positively correlated with the event duration in the internal-perspective task.

The negative correlation between the activation level in the right PPC and sequential distance has already been observed in a previous fMRI study by [Gauthier & van Wassenhove \(2016b\)](#). In their study, the participants were instructed to mentally position themselves at a specific time point and judge whether a target event occurred before or after that time point. The authors identified a similar brain region (reported MNI coordinates of the peak voxel: 42, -70, 40), closely matching the activation observed in the present study (MNI coordinates of the peak voxel: 39, -70, 35). In both studies, activation in this region increased as the target event approached the self-positioned time point, which aligns with the evidence suggesting that the posterior parietal cortex implements egocentric representations. For example, neuropsychological studies have demonstrated that patients with lesions in the bilateral or unilateral right PPC have “egocentric disorientation” ([Aguirre & D’Esposito, 1999](#)): they are unable to localize objects in relation with themselves (e.g., Case 2: [Levine et al., 1985](#); Patient DW: [Stark, 1996](#); Patient MU: [Wilson et al., 1997, 2005](#)). Consistently, we found in a recent fMRI experiment that the distributed activity pattern in the bilateral PPC could encode egocentric but not allocentric directions of the target objects during memory retrieval ([Dutriaux et al., 2024](#)).

What is novel here is that the correlation between the activation level of the right PPC and sequential distance was reversed to a positive one when an external perspective was used. Since an external perspective is often associated with the allocentric reference frame (e.g., [O’Keefe &](#)

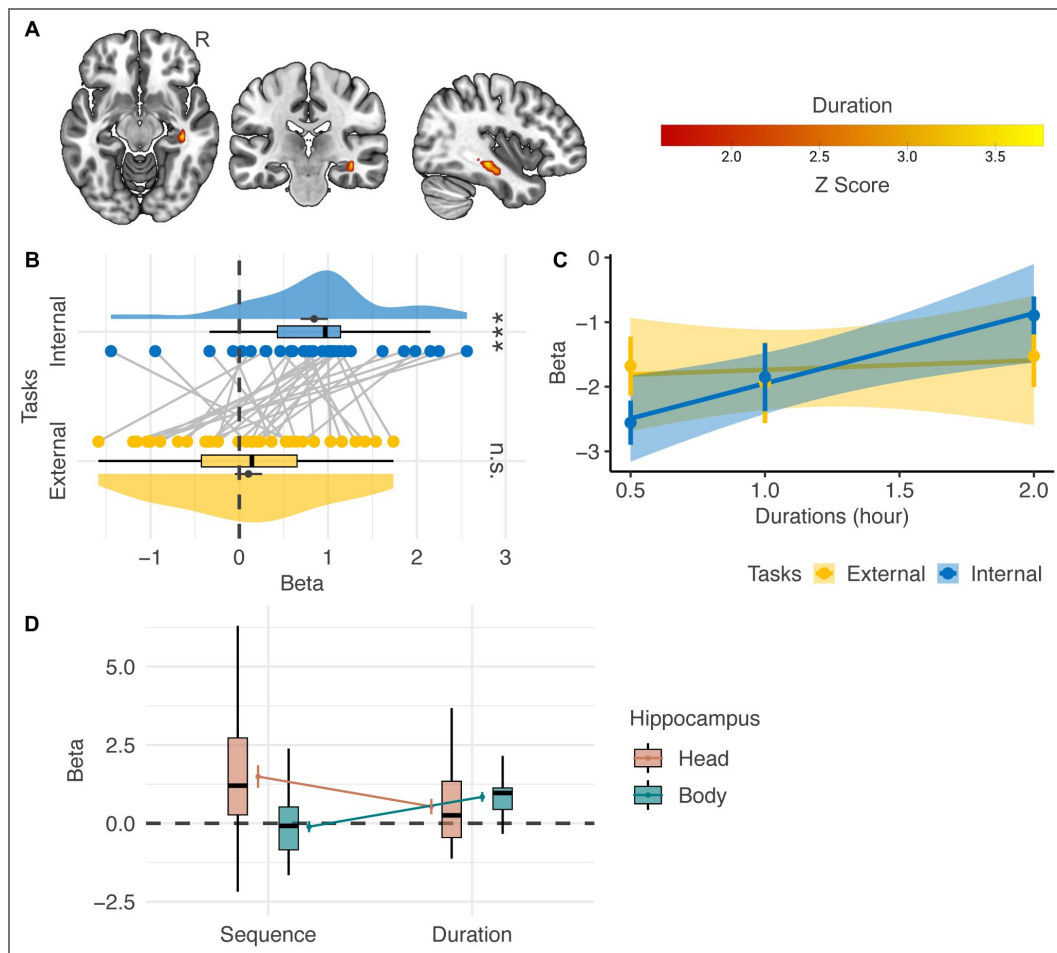


Figure 5. Neural correlates of event duration.

(A) The right hippocampal body shows a significant main effect of *Duration* within the mask of the bilateral hippocampus (voxel-level $p < 0.001$, cluster-level FWE corrected $p < 0.05$; voxel-level $p < 0.05$ for illustration purposes). (B) However, regions of interest analysis shows that the correlation between the activation level in the right hippocampal body and *Duration* significantly differs in the internal- and the external-perspective task. (C) A further illustration of the relations between the activation level in the right hippocampal body and duration in the two tasks. The error bar indicates the standard error relative to the mean, and the shaded band around the linear regression line indicates 95% confidence interval. (D) Directly comparing the effects of *Sequential Distance* and *Duration* in the head and the body of the hippocampus shows a double dissociation pattern: the hippocampal head represented *Sequential Distance* but not *Duration*, while the hippocampal body represented *Duration* but not *Sequential Distance*. R = Right Hemisphere. ***: $p < 0.001$; n.s.: not significant.

Nadel, 1978 [\[1\]](#); Arzy & Schacter, 2019 [\[2\]](#)), our results seem to challenge the view that the parietal cortex implements egocentric representation. However, the conflict is more apparent than real. Perspective-taking, whether internal or external, by definition, must be egocentric, as mental images must be constructed from a specific viewpoint (see similar arguments by Vogele & Fink, 2003 [\[3\]](#); Filimon, 2015 [\[4\]](#)). Therefore, a brain region that implements egocentric representation should vary its activity with the perspectives taken, as shown in the right PPC. This finding is crucial; it suggests that the distance coding in the right PPC is not a “perspective agnostic” magnitude effect, a possibility that previous fMRI studies could not definitively rule out.

Several previous studies have already questioned the nature of the PPC’s function (e.g., see reviews by Andersen et al., 1997 [\[5\]](#); Cohen & Andersen, 2002 [\[6\]](#); Wagner et al., 2005 [\[7\]](#); Abrahamse et al., 2014 [\[8\]](#); Ciaramelli et al., 2008 [\[9\]](#); Cabeza et al., 2008 [\[10\]](#); Hutchinson et al., 2009 [\[11\]](#); Whitlock, 2017 [\[12\]](#); Sestieri et al., 2017 [\[13\]](#); Summerfield et al., 2020 [\[14\]](#); Bottini & Doeller, 2020 [\[15\]](#)). Here, we want to highlight three crucial findings. First, neuropsychological studies indicate that lesions in the bilateral PPC or lateral occipitoparietal cortex will lead to not only egocentric disorientation but also simultanagnosia, one of the key components of Bálint’s syndrome (Bálint, 1909 [\[16\]](#); Rizzo & Vecera, 2001; Chechlacz & Humphreys, 2014 [\[17\]](#); e.g., WF: Holmes and Horax, 1919 [\[18\]](#); MVV: Kase et al., 1977 [\[19\]](#); MU: Wilson et al., 1997, 2005). Patients with simultanagnosia cannot perceive multiple entities as a whole and comprehend the overall meaning of a scene. Second, electrophysiological recordings in the lateral intraparietal area of the macaque cortex find the gain-field neurons, which can be used to transform the reference frames anchored to different body parts (e.g., Andersen et al., 1985 [\[20\]](#); Zipser & Andersen, 1988 [\[21\]](#); Andersen et al., 1997 [\[5\]](#); Cohen & Andersen, 2002 [\[6\]](#)). Third, the fMRI studies in humans show that the particular PPC area found in our study (i.e., the HIP5 area in the lateral wall of the intraparietal sulcus and the dorsal PGp in the angular gyrus) are more engaged in episodic memory retrieval, in contrast to the medial wall of the intraparietal sulcus and dorsal part of the PPC more involved in perceptual attention (e.g., Hutchinson et al., 2009 [\[11\]](#); Sestieri et al., 2010 [\[22\]](#); Sestieri et al., 2017 [\[13\]](#)). Taking all these findings together, the PPC area identified in this study might contribute to memory retrieval in egocentric reference frames, which maintains relations among multiple memorized entities in the working memory from a perspective optimal for the current task context (Abrahamse et al., 2014 [\[8\]](#)). Such relational schema, constructed from stereotypical perspectives, is also represented in the PPC as a template to guide attention (Summerfield et al., 2020 [\[14\]](#); Bottini & Doeller, 2020 [\[15\]](#)). It is thus not surprising that the two most common perspectives for time construals—mental time travel and mental time watching—involve the PPC.

In contrast to the PPC, we observed that the activation level in the head of the right hippocampus positively correlated with the sequential distance, regardless of the perspectives. Previous studies have already shown that the hippocampal activation level correlates with distance (e.g., Morgan et al., 2011; Howard et al., 2014; Garvert et al., 2017 [\[23\]](#); Theves et al., 2019 [\[24\]](#); Viganò et al., 2023 [\[25\]](#)), and the distributed activity in the hippocampus can encode distance (e.g., Deuker et al., 2016 [\[26\]](#); Park et al., 2021). Most studies reported hippocampal effects either bilaterally or predominantly in the right hemisphere, whereas only one study identified the effect in the left hippocampus instead (i.e., Morgan et al., 2011). Our study is novel in showing that this distance coding in the hippocampus is independent of perspectives, indicating an allocentric representation of the event series. This finding aligns with the hypothesis in the spatial domain suggesting the hippocampus as an allocentric cognitive map (e.g., O’Keefe & Nadel, 1978 [\[27\]](#); Byrne et al., 2007 [\[28\]](#); Bicanski & Burgess, 2018 [\[29\]](#); Bicanski & Burgess, 2020 [\[30\]](#); Bottini & Doeller, 2020 [\[15\]](#)). The positive correlation between the sequential distance and the hippocampal activation level might be mediated by adaptation (e.g., Grill-Spector & Malach, 2001): the longer the distance, the less the adaptation, and the greater the hippocampal activation. One way to interpret such perspective-agnostic representation in the hippocampus is to view it as the associations among memorized entities (e.g., Muller et al., 1996 [\[31\]](#); Eichenbaum, 2014 [\[32\]](#), 2017 [\[33\]](#); Buzsáki & Tingley, 2018 [\[34\]](#); Quiroga, 2019). As a result, the allocentric representation in the hippocampus may not require a “reference frame” because there is no fixed reference point. Each entity is represented among the relations to others, contrasting with the egocentric representation in the parietal cortex, where the reference point is clearly the self.

In this context, the distinction between the allocentric and the egocentric representations of an event series can also be understood in terms of memory storage and retrieval (also see Byrne et al., 2007). The hippocampus stores event series in a static and perspective-agnostic manner, while the PPC flexibly retrieves memory by constructing egocentric images tied to variable perspectives. Supporting this hypothesis is evidence that bilateral hippocampal lesions result in memory deficit for event series (e.g., Dede et al., 2016), whereas bilateral PPC lesions impair free recall but do not cause memory loss (e.g., Berryhill et al., 2007). Patients can still recall details of past events when prompted with cues or questions but struggle to access memories spontaneously.

Finally, the study indicates that the event durations were represented in the right hippocampal body, in line with previous studies suggesting that the hippocampus contributes to representing durations that are embedded within the event sequence (e.g., Barnett et al., 2014; Thavabalasingam et al., 2018; Lee et al., 2020). Notably, the post hoc analysis reveals a double dissociation pattern: the hippocampal head represented sequential distances between events, whereas the hippocampal body represented the duration of individual events. Evidence from the spatial domain has suggested that the anterior hippocampus (or the ventral rodent hippocampus) implements global and gist-like representations (e.g., larger receptive fields), whereas the posterior hippocampus (or the dorsal rodent hippocampus) implements local and detailed ones (e.g., finer receptive fields) (e.g., Jung et al., 1994; Kjelstrup et al., 2008; Collin et al., 2015; see reviews by Poppenk et al., 2013; Robin & Moscovitch, 2017; see Strange et al., 2014 for a different opinion). Recent evidence further shows that the organizational principle observed along the hippocampal long axis may also extend to the temporal domain (Montagrin et al., 2024). In that study, the anterior hippocampus showed greater activation for remote goals, whereas the posterior hippocampus was more strongly engaged for current goals, which are presumed to be represented in finer detail. Thus, the hippocampus likely represents the sequence of events hierarchically along its longitudinal axis: from the gist event sequence to the event sequence constituting a gist event, down to the unitary time bin sequence constituting an event as its duration. This view unifies the hippocampus's role in sequence representation and explains why it only represents durations embedded within event sequences. Future studies are required to confirm this hypothesis and investigate whether such hierarchical structures along the longitude axis of the hippocampus can also be used to represent the events at different time scales (e.g., from years to seconds). It would also be intriguing to examine whether the parietal cortex has a similar hierarchical structure or plays a role in zoom-in and zoom-out events at different scales to maintain an appropriate working memory load.

To conclude, this study reveals the neural correlates of sequence and duration in time construals. The hippocampal head represents the event sequence allocentrically, irrespective of the observer's perspectives, whereas the hippocampal body represents the event duration embedded in the event sequence. The posterior parietal cortex flexibly constructs the egocentric representation of the event series that varies according to the observer's perspectives, which could be internal (i.e., mental time travel) or external (i.e., mental time watching). Such allocentric and egocentric representations of event series can be interpreted in terms of memory storage and retrieval.

Methods

Participants

Thirty-five native Italian speakers with no history of neurobiological or psychiatric disorders participated in the experiment. The sample size was chosen to align with the upper range of participant numbers reported in previous fMRI studies that successfully detected sequence or distance effects in the hippocampus ($N = 15\text{--}34$; e.g., Morgan et al., 2011; Howard et al., 2014; Deuker et al., 2016; Garvert et al., 2017; Theves et al., 2019; Park et al., 2021; Cristoforetti et al., 2022). The ethical committee of the University of Trento approved the experimental protocol (Approval Number 2019-018), and all participants provided written informed consent and were paid for their time. Three participants were excluded. Two were due to poor behavioral performance during scanning; their accuracy was below 1.5 times the interquartile range below

the lower quartile across participants. One was due to excessive head motion during scanning; this participant's mean frame displacement index (Power et al., 2014) was above 1.5 times the interquartile range above the upper quartile across participants. The remaining 32 participants entered the analysis (19 females, 13 males; age: 19-34; $M = 24.0$; $SD = 3.8$; all participants except one were right-handed).

Stimuli

We created a fictional religious ritual as the stimulus. Like most rituals, it follows a specific sequence, endures particular durations, and happens on predetermined parts of the day (Figure 1A). This created ritual comprised 15 events, falling into three parts of the day, i.e., morning, afternoon, and evening. Each part of the day included five events and lasted six hours in total: two events lasted for half an hour, one for one hour, and the other two for two hours. We instantiated each event with an event phrase. To minimize the potential confounding between the time information of the events and the semantic information of the phrases, we randomly assigned 15 phrases to the events twice, generating two versions for even and odd numbers of participants.

Procedures

The day before fMRI scanning, participants learned the temporal information of the ritual: the sequence, the durations, and the parts of the day. The learning procedure included two phases. The first was the reading phase, and the second was the imagination phase. The two phases combined were performed twice.

In the reading phase, participants read a narrative describing the whole ritual on a computer screen twice. The computer screen presented one sentence at a time, and each sentence provided information on one event. The presentation sequence was identical to the event sequence of the fictional ritual, and the sentences described the duration and the parts of the day of each event. Participants read the sentences at their own pace by pressing the spacebar to read the following sentence.

In the imagination phase, participants imagined themselves performing the whole ritual one event after another, guided by the prerecorded auditory instructions delivered through the headphones. Each event started with a voice telling the event to be imagined (e.g., "light some candle"). A single beep was then played to indicate the start of the imagination, and a double beep sound indicated the end of the imagination. The imagining sequence was the same as the actual sequence, the imagining duration (i.e., the interval between the single and the double beep) was proportional to the actual duration (30 seconds / 1 hour), and the parts of the day were indicated in the instruction (e.g., "the morning starts"). Participants were told to close their eyes to avoid distractions and imagined the whole ritual four times in each imagining phase (i.e., eight times in total).

After learning, participants' knowledge of the ritual was assessed with three tests. In the event-sequence test, participants judged whether one event happened in the past or the future of another event. In the event-duration test, participants imagined the ritual in a self-paced manner and pressed the button when finishing the imagination of each event. In the parts-of-the-day test, participants judged whether two events happened in the same or different parts of the day. All participants' performance was greater than 80% in both the event-sequence and parts-of-the-day tests. In the event-duration test, Pearson's correlation between the self-paced imagining duration and the actual duration across 15 events was greater than 0.6 in all participants.

The scanning consisted of six runs, each with one external-perspective task block and one internal-perspective task block. The order of these two task blocks was interleaved across the six runs within each participant, and the order of these two task blocks in the first run was counterbalanced across participants. That means half the participants followed the order "EI-IE-EI-IE-EI-IE", and half of the participants followed the order "IE-EI-IE-EI-IE-EI" ("E" indicating the external-perspective task block and "I" indicating the internal-perspective task block). Each task block started with a 5s task prompt indicating the task of this block and a 3s countdown presentation (i.e., the screen presenting "3", "2", "1"). The task block had 20 trials that were identical in the two task blocks within the same run but were presented in a randomized order.

In each trial, the phrases of two events were visually presented one after the other (Figure 1C). Participants first saw a 0.5s fixation cross, the phrase of the reference event for 0.8s, and a blank screen for 3.2s (75% trials) or 7.2s (25% trials). They then saw another 0.5s fixation cross and the phrase of the target event. Each event phrase was presented in the center of a grey screen in two rows (black color, Calibri font): in the first row was the verb of the phrase (i.e., “light”), and in the second row was the object of this verb “some candle”). When presenting the target event phrase, we also presented two possible answers under the target event phrase; they were smaller in font size and colored differently depending on the task (i.e., red for the external-perspective task and blue for the internal-perspective task). In the external-perspective task block, the participants judged whether the target event happened in the same or different part of the day of the reference event, and the two option words were “same” and “different”. In the internal-perspective task block, the participants were instructed to project themselves into the reference event and judge whether the target event happened in the past or will happen in the future, and the two option words were “past” and “future”.

Participants were instructed to perform the two tasks once they saw the target event phrase by pressing two buttons with their right index and middle fingers, corresponding to the left and the right options on the screen, respectively. The reaction time was calculated as the duration between the onset of the target event phrase and the button press. A blank screen would replace the target event presentation once the response was given, and the total duration of the target event and the blank screen was maintained as 4s. On 75% of occasions, the subsequent trial started immediately; On the other 25%, there would be another 4s blank screen between trials.

We carefully selected two sets of 20 event pairs from the 210 possible combinations, assigning them to the odd and even runs of the fMRI experiment. Using a brute-force search, we identified 20 pairs in which sequential distance showed only weak correlations with positional information for both reference and target events (ranging from 1 to 15), as well as with behavioral responses (Same vs. Different and Future vs. Past, coded as 0 and 1), with all correlation coefficients below 0.2. At the same time, we balanced the proportion of correct responses across conditions: for the external-perspective task, Same/Different = 11/9 and 12/8; for the internal-perspective task, Future/Past = 12/8 and 8/12. Under these constraints, the sequential distances in both sets ranged from 1 to 5. To further mitigate spatial response biases, we pseudorandomized the left/right on-screen positions of the two response options within each task block, while ensuring an equal number of correct responses mapped to the left and right buttons (i.e., 10 per block).

Behavior analysis

To investigate the factors affecting trial-by-trial RT of the correct trials, we built linear mixed models (LMM) with *Participant* as the random effects grouping factor using the lme4 package (Bates et al., 2015). We fit the maximal model including the random intercept and all the random slopes consistent with the experimental design (as recommended by Barr et al., 2013). In the case of overfitting (singular fit), we removed the random slopes but kept the random intercept (Matuschek et al., 2017).

MRI acquisition

MRI data were acquired using a MAGNETOM Prisma 3T MR scanner (Siemens) with a 64-channel head-neck coil at the Centre for Mind/Brain Sciences, University of Trento. Functional images were acquired using the simultaneous multislice echoplanar imaging sequence: the scanning plane was parallel to the long axis of the hippocampus, the phase encoding direction was from anterior to posterior, repetition time (TR) = 1000 ms, echo time (TE) = 28 ms, flip angle (FA) = 59°, field of view (FOV) = 200 mm × 200 mm, matrix size = 66 × 66, 65 axial slices, slices thickness (ST) = 3 mm, no gap, voxel size = 3.03 × 3.03 × 3 mm, multiband factor = 5.

Field maps were acquired between each pair of functional runs to correct for geometric distortions of these two runs. Each set of field maps included three images, one phase-drift image between two slightly different echo times and two magnitude images for each of these two echo times: the scanning plane was parallel to the long axis of the hippocampus, the phase encoding

direction was from anterior to posterior, TR = 1030 ms, shorter TE = 4.92 ms, longer TE = 7.38 ms, FA = 60°, FOV = 210 mm × 210 mm, matrix size = 70 × 70, 66 axial slices, ST = 3 mm, no gap, voxel size = 3 × 3 × 3 mm.

Three-dimensional T1-weighted images were acquired using the magnetization-prepared rapid gradient-echo sequence, sagittal plane, TR = 2530 ms, TE = 1.69 ms, inversion time = 1100 ms, FA = 7°, FOV = 256 mm × 256 mm, matrix size = 256 × 256, 176 continuous sagittal slices, ST = 1 mm, voxel size = 1 × 1 × 1 mm.

MRI preprocessing

We preprocessed the brain images using SPM12

(<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>). The functional images were first realigned to the first image in the first run, generating six rigid head motion parameters for each time point and a mean functional image across all the runs. We then used the field maps to calculate the voxel displacement maps and coregistered them to this mean functional image. Since we acquired the field maps between each pair of functional runs, we applied each voxel displacement map to its two closest functional runs to correct their geometric distortions. The resulting functional images were next normalized to the MNI space with the acquired T1-weighted image using the unified segmentation method. In the final step, we spatially smoothed the normalized functional images, and the full width at half maximum of the 3D Gaussian smoothing kernel was 4 mm.

First-level fMRI analysis

We performed the first-level analysis using SPM12

(<https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>). General linear models (GLMs) were built to predict each participant's blood-oxygen-level-dependent (BOLD) signal. The GLMs in the primary analysis included six conditions: the task prompt and the countdown at the beginning of each block, the reference event and the target event in the external-perspective task, and the reference event and the target event in the internal-perspective task. The duration of the task prompt and the countdown were set as the duration of the actual presentation (i.e., 5s and 3s, respectively). The duration of the four event conditions was set as 0. The resulting boxcar and stick functions were convolved with a canonical hemodynamic response function (HRF). Head motion parameters and constant variables indicating each of the six runs were included as nuisance regressors. A high-pass filter with a cutoff of 128s was used to remove the low-frequency noise and slow signal drifts.

We used the parametric modulation method to investigate the two core temporal variables: sequence (i.e., the number of events between the reference event and the target event) and duration (i.e., the duration of the target events in terms of hours). The z-scores of these parameters across the events in each run were set as parameters modulating the target events in both external- and internal-perspective tasks. The option for orthogonalizing modulations in the SPM was turned off (Mumford et al., 2015).

As a sanity check, we investigated whether the parametric modulation method in this study can successfully detect the visual cortex whose activation should be modulated by the number of syllables of the visually presented event phrase. To this end, we built a stick function with the sticks located on the onsets of both reference and target events and the “height” of the stick as the z score of the number of syllables across the event phrases in each run. This stick function was convolved with a canonical HRF as an additional regressor involved in the GLM.

To detect the specific activations for external- and internal-perspective tasks, we directly contrasted the target event in the external-perspective task and the target event in the internal-perspective task. To investigate how the neural activation was modulated by each of the temporal information across different perspectives, we looked at its interaction effect with the task (i.e., contrast the effect between external- and internal-perspective tasks, i.e., contrast weights vector: 1, -1) and the main effect regardless of tasks (i.e., contrast weights vector: 1, 1).

To validate whether any significant effects were explained by the RT, we also built the same GLM incorporating trial-by-trial RT as a covariate. To do so, we created a stick function with the sticks positioned at the onset of the target events in both external- and internal-perspective tasks, and the height of each stick was set to the z-score of the corresponding RT. These sticks were convolved with a canonical HRF to serve as a regressor in the GLM.

To directly illustrate how the activation level varied with the sequential distance, we built an independent GLM with each distance (i.e., from 1 to 5) in each task as a separate condition (i.e., ten conditions in total). To directly illustrate how the activation level varied with duration, we built another GLM with each duration (i.e., 0.5, 1, and 2 hours) in each task as a separate condition (i.e., six conditions in total). In both GLMs, we involved the task prompt and the countdown as conditions of no interest. The duration of the task prompt and the countdown were set as the duration of the actual presentation (i.e., 5s and 3s, respectively). The duration of all the other conditions was set as 0. The resulting boxcar and stick functions were convolved with the canonical HRF. Head motion parameters and constant variables indicating each of the six runs were included as nuisance regressors. A high-pass filter with a cutoff of 128s was used to remove the low-frequency noise and slow signal drifts.

Second-level fMRI analysis

We performed the group-level one-sample test on the first-level beta images from the univariate contrast and parametric modulation analyses. The statistical inference was performed using the permutation method with PALM toolbox (<https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/PALM>). Five thousand sign flips were performed (Winkler et al., 2014), and a generalized Pareto distribution was fit to model the tail of the permutation distribution for the P values below 0.1 (Knijnenburg et al., 2009; Winkler et al., 2016). We controlled the family-wise error rate using a conventional cluster-forming threshold (i.e., voxel-wise $p < 0.001$, cluster-level FWE corrected $p < 0.05$) (Woo et al., 2014). Given that the hippocampus is our primary region of interest, its elongated and thin shape limits the number of contiguous voxels, restricting the formation of large clusters, and functionally independent clusters within the hippocampus may naturally be small, we performed multiple comparison corrections separately for the cortex and the hippocampus. We used the Automated Anatomical Labelling Atlas 3 to define these masks (Rolls et al., 2020). The cortical mask was defined as all the cortical regions in both hemispheres combined, and the hippocampal mask was defined as the hippocampus in both hemispheres combined (i.e., the No.41 and the No. 42 areas combined).

Figure supplements

Parts of the Day	Sequence	Duration (Hours)	Event Phrase	
			Odd Number Participants	Even Number Participants
Morning	1	1	light some candles	dance the sacred dance
	2	0.5	recite the prayers	prepare the sacred cake
	3	2	ring the bell	talk to a priest
	4	0.5	make an origami	practice breathing
	5	2	dive into the water	visit the temple
Afternoon	6	0.5	worship the relic	meditate on life
	7	2	write some resolutions	recite the prayers
	8	2	sing sacred hymns	plant an oak
	9	1	meditate on life	sing sacred hymns
	10	0.5	talk to a priest	write some resolutions
Evening	11	2	plant an oak	make an origami
	12	0.5	practice breathing	worship the relic
	13	2	prepare the sacred cake	light some candles
	14	0.5	visit the temple	dive into the water
	15	1	dance the sacred dance	ring the bell

Figure 1—figure supplement 1. All participants learnt the same event structure. However, the event order differed across participants (i.e., two event lists were used for even- and odd-numbered participants). This approach helps minimize potential confounding between the temporal and semantic information carried by the events.

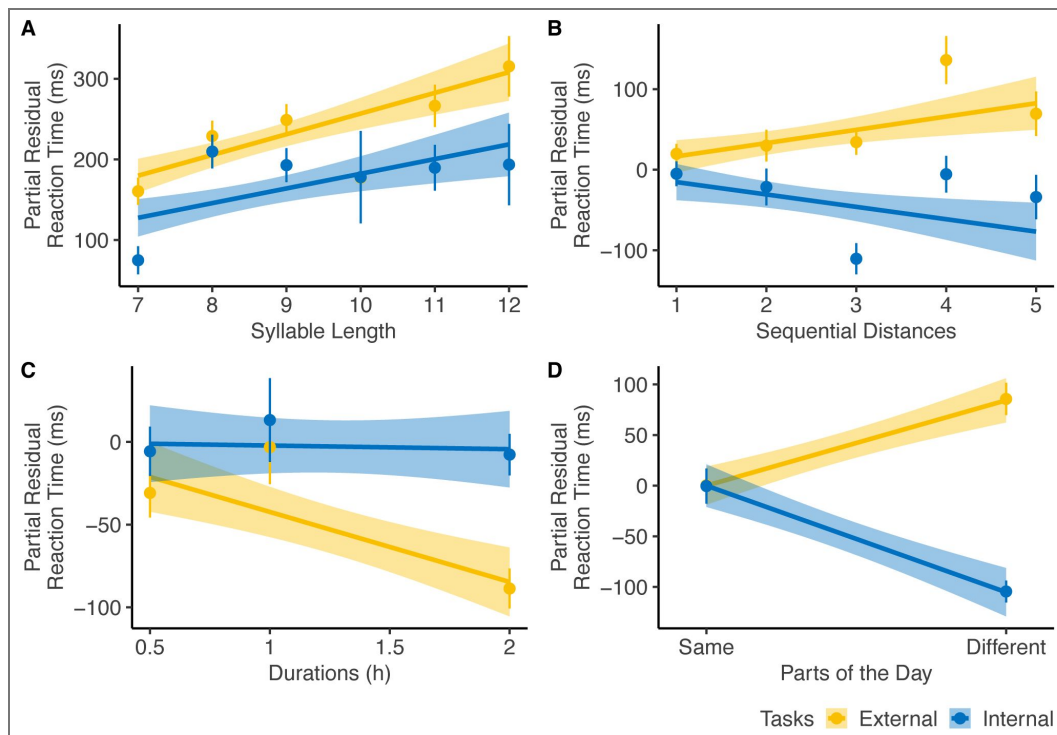


Figure 2—figure supplement 1. Reaction time analysis including *Parts of the Day*.

For diagnostic purposes, we plotted the partial residuals of each significant predictor significantly influencing the reaction time (RT) of the corrected trials. The partial residual includes the effect of each variable, its interaction with *Task Type*, and the residuals from the full linear regression model. A. RT increased with *Syllable Length*, showing a similar trend across both tasks. B. *Sequential Distance* affected RT in opposite directions depending on perspectives. C. *Event Duration* influenced RT only in the external-perspective task, with no effect in the internal-perspective task. D. The effect of *Parts of the Day* was in line with the effect of *Sequential Distance*, with two events in the same and different parts of the day corresponding to the short and long sequential distances, respectively.

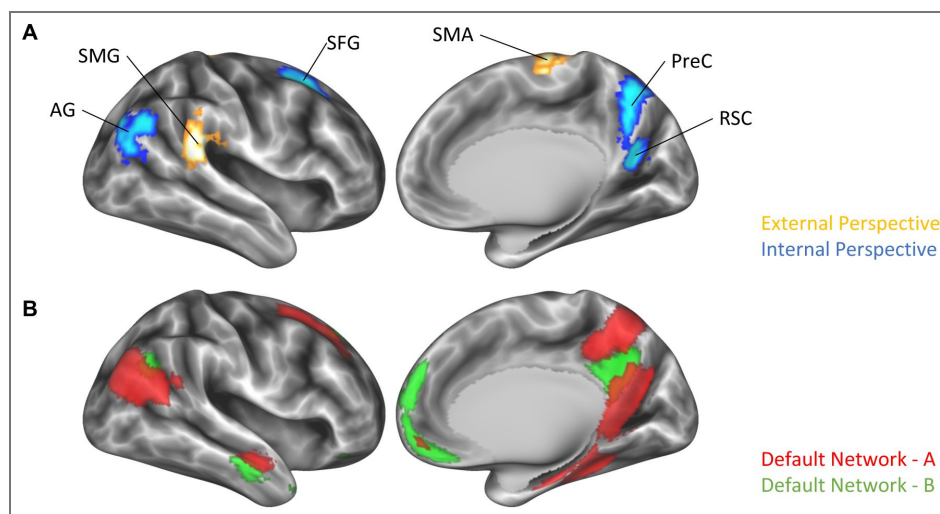


Figure 3—figure supplement 1.

(A) Brain surface view of the univariate contrast between external-perspective and internal-perspective tasks (voxel-level $p < 0.001$, cluster-level FWE corrected $p < 0.05$). This view is transformed from the significant clusters in the MNI space in Figure 3A to the fsLR space using the toolbox neuromaps (<https://github.com/netneurolab/neuromaps>). All the significant areas were in the right hemisphere. (B) Brain surface view of the default network A and the default network B. They were respectively identified as the 8th and the 1st components among the 25 components of the “group-ICA” template from the UK Biobank brain imaging (<https://www.fmrrib.ox.ac.uk/ukbiobank/>). We preserved the positive voxels above 7 in the MNI space and transformed them into the fsLR space using the toolbox neuromaps. Both plots are illustrated using the Connectome Workbench 2.0 (<https://www.humanconnectome.org/software/connectome-workbench>). They are displayed on an inflated surface against the group-averaged all sulcus image of 1096 young adults from the dataset of the Human Connectome Project (<https://balsa.wustl.edu/reference/pkXDZ>).

Data availability

Figure 2-source data 1, Figure 3-source data 1, Figure 4-source data 1, and Figure 5-source data 1 contain the data used to generate the corresponding figures.

Acknowledgements

This work was supported by the European Research Council (ERC-StG, NOAM 804422) and the Italian Ministry of University and Research (MUR-FARE, MODGET R18WJMSNZF) attributed to R.B. We thank Mattia Silvestri for the help with data collection and Simone Viganò for the discussion.

Additional files

[Supplementary File 1](#) 

[Figure 1-source data 1](#) 

[Figure 2-source data 1](#) 

[Figure 3-source data 1](#) 

[Figure 4-source data 1](#) 

[Figure 5-source data 1](#) 

Additional information

Funding

Funder	Grant reference number	Author
EC European Research Council (ERC)	804422	Roberto Bottini
Italian Ministry of University and Research	R18WJMSNZF	Roberto Bottini

Author ORCID iDs

Yangwen Xu: <https://orcid.org/0000-0003-1482-4927>

References

- Abrahamse E., Van Dijk J. P., Majerus S., Fias W. (2014) Finding the answer in space: The mental whiteboard hypothesis on serial order in working memory. *Frontiers in Human Neuroscience* **8**:932
- Aguirre G. K., D'Esposito M (1999) Topographical disorientation: a synthesis and taxonomy. *Brain* **122**:1613-1628
- Andersen R. A., Essick G. K., Siegel R. M (1985) Encoding of spatial location by posterior parietal neurons. *Science* **230**:456-458
- Andersen R. A., Snyder L. H., Bradley D. C., Xing J (1997) Multimodal representation of space in the posterior parietal cortex and its use in planning movements. *Annual Review of Neuroscience* **20**:303-330
- Arzy S., Schacter D. L (2019) Self-agency and self-ownership in cognitive mapping. *Trends in Cognitive Sciences* **23**:476-487
- Arzy S., Adi-Japha E., Blanke O (2009a) The mental time line: An analogue of the mental number line in the mapping of life events. *Consciousness and Cognition* **18**:781-785
- Arzy S., Collette S., Ionta S., Fornari E., Blanke O (2009b) Subjective mental time: the functional architecture of projecting the self to past and future. *European Journal of Neuroscience* **30**:2009-2017

- Arzy S., Molnar-Szakacs I., Blanke O (2008) Self in time: imagined self-location influences neural activity related to mental time travel. *Journal of Neuroscience* **28**:6502-6507
- Bálint R. (1909) Seelenlähmung des "Schauens", optische Ataxie, räumliche Störung der Aufmerksamkeit. pp. 51–66. *European Neurology* **25**:51-66
- Barnett A. J., O'Neill E. B., Watson H. C., Lee A. C. (2014) The human hippocampus is sensitive to the durations of events and intervals within a sequence. *Neuropsychologia* **64**:1-12
- Barr D. J., Levy R., Scheepers C., Tily H. J (2013) Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language* **68**:255-278
- Bates D., Mächler M., Bolker B., Walker S (2015) Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* **67**:1-48
- Bellmund J. L., Deuker L., Doeller C. F (2019) Mapping sequence structure in the human lateral entorhinal cortex. *eLife* **8**:e45333 <https://doi.org/10.7554/eLife.45333>
- Bellmund J. L., Deuker L., Montijn N. D., Doeller C. F (2022) Mnemonic construction and representation of temporal structure in the hippocampal formation. *Nature Communications* **13**:3395
- Bellmund J. L., Polti I., Doeller C. F (2020) Sequence memory in the hippocampal-entorhinal region. *Journal of Cognitive Neuroscience* **32**:2056-2070
- Bender A., Beller S (2014) Mapping spatial frames of reference onto time: A review of theoretical accounts and empirical findings. *Cognition* **132**:342-382
- Berryhill M. E., Phuong L., Picasso L., Cabeza R., Olson I. R (2007) Parietal lobe and episodic memory: bilateral damage causes impaired free recall of autobiographical memory. *Journal of Neuroscience* **27**:14415-14423
- Bicanski A., Burgess N (2018) A neural-level model of spatial memory and imagery. *eLife* **7**:e33752 <https://doi.org/10.7554/eLife.33752>
- Bicanski A., Burgess N. (2020) Neuronal vector coding in spatial cognition. *Nature Reviews Neuroscience* **21**:453-470
- Blanke O., Arzy S (2005) The out-of-body experience: disturbed self-processing at the temporo-parietal junction. *The Neuroscientist* **11**:16-24
- Blanke O., Landis T., Spinelli L., Seeck M (2004) Out-of-body experience and autoscapy of neurological origin. *Brain* **127**:243-258
- Bottini R., Doeller C. F (2020) Knowledge across reference frames: Cognitive maps and image spaces. *Trends in Cognitive Sciences* **24**:606-619
- Braga R. M., Buckner R. L (2017) Parallel interdigitated distributed networks within the individual estimated by intrinsic functional connectivity. *Neuron* **95**:457-471
- Buckner R. L., Carroll D. C (2007) Self-projection and the brain. *Trends in Cognitive Sciences* **11**:49-57
- Buonomano D. (2017) *Your brain is a time machine: The neuroscience and physics of time* WW Norton & Company.
- Buonomano D. V., Laje R (2010) Population clocks: motor timing with neural dynamics. *Trends in Cognitive Sciences* **14**:520-527
- Buzsáki G., Tingley D (2018) Space and time: the hippocampus as a sequence generator. *Trends in Cognitive Sciences* **22**:853-869
- Byrne P., Becker S., Burgess N (2007) Remembering the past and imagining the future: a neural model of spatial memory and imagery. *Psychological Review* **114**:340
- Cabeza R., Ciaramelli E., Olson I. R., Moscovitch M (2008) The parietal cortex and episodic memory: an attentional account. *Nature Reviews Neuroscience* **9**:613-625
- Caspers S., Eickhoff S. B., Geyer S., Scheperjans F., Mohlberg H., Zilles K., Amunts K (2008) The human inferior parietal lobule in stereotaxic space. *Brain Structure and Function* **212**:481-495

- Caspers S., Geyer S., Schleicher A., Mohlberg H., Amunts K., Zilles K (2006) The human inferior parietal cortex: cytoarchitectonic parcellation and interindividual variability. *Neuroimage* **33**:430-448
- Chechlacz M., Humphreys G. W (2014) The enigma of Bálint's syndrome: neural substrates and cognitive deficits. *Frontiers in Human Neuroscience* **8**:123
- Ciaramelli E., Grady C. L., Moscovitch M (2008) Top-down and bottom-up attention to memory: a hypothesis (AtoM) on the role of the posterior parietal cortex in memory retrieval. *Neuropsychologia* **46**:1828-1851
- Cohen Y. E., Andersen R. A (2002) A common reference frame for movement plans in the posterior parietal cortex. *Nature Reviews Neuroscience* **3**:553-562
- Cohn-Sheehy B. I., Ranganath C (2017) Time regained: how the human brain constructs memory for time. *Current Opinion in Behavioral Sciences* **17**:169-177
- Collin S. H., Milivojevic B., Doeller C. F (2015) Memory hierarchies map onto the hippocampal long axis in humans. *Nature Neuroscience* **18**:1562-1564
- Cristoforetti G., Majerus S., Sahan M. I., Van Dijk J. P., Fias W. (2022) Neural patterns in parietal cortex and hippocampus distinguish retrieval of start versus end positions in working memory. *Journal of Cognitive Neuroscience* **34**:1230-1245
- De Kock R., Gladhill K. A., Ali M. N., Joiner W. M., Wiener M. (2021) How movements shape the perception of time. *Trends in Cognitive Sciences* **25**:950-963
- Dede A. J., Frascino J. C., Wixted J. T., Squire L. R (2016) Learning and remembering real-world events after medial temporal lobe damage. *Proceedings of the National Academy of Sciences* **113**:13480-13485
- Deuker L., Bellmund J. L., Navarro Schröder T., Doeller C. F (2016) An event map of memory space in the hippocampus. *eLife* **5**:e16534 <https://doi.org/10.7554/eLife.16534>
- DiNicola L. M., Braga R. M., Buckner R. L (2020) Parallel distributed networks dissociate episodic and social functions within the individual. *Journal of Neurophysiology* **123**:1144-1179
- Dutriaux L., Xu Y., Sartorato N., Lhuillier S., Bottini R (2024) Disentangling reference frames in the neural compass. *Imaging Neuroscience* **2**:1-18
- Eichenbaum H (2014) Time cells in the hippocampus: a new dimension for mapping memories. *Nature Reviews Neuroscience* **15**:732-744
- Eichenbaum H (2017) On the integration of space, time, and memory. *Neuron* **95**:1007-1018
- Eickhoff S. B., Stephan K. E., Mohlberg H., Grefkes C., Fink G. R., Amunts K., Zilles K. (2005) A new SPM toolbox for combining probabilistic cytoarchitectonic maps and functional imaging data. *Neuroimage* **25**:1325-1335
- Fedorenko E., Duncan J., Kanwisher N (2013) Broad domain generality in focal regions of frontal and parietal cortex. *Proceedings of the National Academy of Sciences* **110**:16616-16621
- Filimon F (2015) Are all spatial reference frames egocentric? Reinterpreting evidence for allocentric, object-centered, or world-centered reference frames. *Frontiers in Human Neuroscience* **9**:648
- Gauthier B., van Wassenhove V. (2016a) Cognitive mapping in mental time travel and mental space navigation. *Cognition* **154**:55-68
- Gauthier B., van Wassenhove V. (2016b) Time is not space: core computations and domain-specific networks for mental travels. *Journal of Neuroscience* **36**:11891-11903
- Gauthier B., Pestke K., van Wassenhove V. (2019) Building the arrow of time... over time: a sequence of brain activity mapping imagined events in time and space. *Cerebral Cortex* **29**:4398-4414
- Gauthier B., Prabhu P., Kotegar K. A., van Wassenhove V. (2020) Hippocampal contribution to ordinal psychological time in the human brain. *Journal of Cognitive Neuroscience* **32**:2071-2086
- Garvert M. M., Dolan R. J., Behrens T. E (2017) A map of abstract relational knowledge in the human hippocampal-entorhinal cortex. *eLife* **6**:e17086 <https://doi.org/10.7554/eLife.17086>

- Holmes G., Horrax G (1919) Disturbances of spatial orientation and visual attention, with loss of stereoscopic vision. *Archives of Neurology & Psychiatry* **1**:385-407
- Hutchinson J. B., Uncapher M. R., Wagner A. D (2009) Posterior parietal cortex and episodic retrieval: convergent and divergent effects of attention and memory. *Learning & Memory* **16**:343-356
- Jung M. W., Wiener S. I., McNaughton B. L (1994) Comparison of spatial firing characteristics of units in dorsal and ventral hippocampus of the rat. *Journal of Neuroscience* **14**:7347-7356
- Kase C. S., Troncoso J. F., Court J. E., Tapia J. F., Mohr J. P (1977) Global spatial disorientation: clinico-pathologic correlations. *Journal of the Neurological Sciences* **34**:267-278
- Kjelstrup K. B., Solstad T., Brun V. H., Hafting T., Leutgeb S., Witter M. P., ..., Moser M. B. (2008) Finite scale of spatial representation in the hippocampus. *Science* **321**:140-1
- Knijnenburg T. A., Wessels L. F., Reinders M. J., Shmulevich I. (2009) Fewer permutations, more accurate P-values. *Bioinformatics* **25**:i161-i168
- Kosslyn S. M., Ball T. M., Reiser B. J (1978) Visual images preserve metric spatial information: evidence from studies of image scanning. *Journal of Experimental Psychology: Human Perception and Performance* **4**:47
- Lakoff G., Johnson M (1980) Conceptual metaphor in everyday language. *Journal of Philosophy* **77**:453-486
- Lee A. C., Thavabalasingam S., Alushaj D., Çavdaroğlu B., Ito R (2020) The hippocampus contributes to temporal duration memory in the context of event sequences: A cross-species perspective. *Neuropsychologia* **137**:107300
- Levine D. N., Warach J., Farah M (1985) Two visual systems in mental imagery: Dissociation of “what” and “where” in imagery disorders due to bilateral posterior cerebral lesions. *Neurology* **35**:1010-1010
- Lin N., Wang X., Xu Y., Wang X., Hua H., Zhao Y., Li X (2018) Fine subdivisions of the semantic network supporting social and sensory-motor semantic processing. *Cerebral Cortex* **28**:2699-2710
- Matuschek H., Kliegl R., Vasishth S., Baayen H., Bates D (2017) Balancing Type I error and power in linear mixed models. *Journal of Memory and Language* **94**:305-315
- McTaggart J. E (1908) The unreality of time. *Mind* 457-474
- Montagrin A., Croote D. E., Preti M. G., Lerman L., Baxter M. G., Schiller D (2024) The hippocampus dissociates present from past and future goals. *Nature Communications* **15**:4815
- Moyer R. S., Bayer R. H. (1976) Mental comparison and the symbolic distance effect. *Cognitive Psychology* **8**:228-246
- Moyer R. S., Landauer T. K (1967) Time required for judgements of numerical inequality. *Nature* **215**:1519-1520
- Muller R. U., Stead M., Pach J (1996) The hippocampus as a cognitive graph. *The Journal of General Physiology* **107**:663
- Mumford J. A., Poline J. B., Poldrack R. A (2015) Orthogonalization of regressors in fMRI models. *PLOS One* **10**:e0126255
- Mumford J. A., Bissett P. G., Jones H. M., Shim S., Rios J. A. H., Poldrack R. A (2024) The response time paradox in functional magnetic resonance imaging analyses. *Nature Human Behaviour* **8**:349-360
- Nani A., Manuella J., Liloia D., Duca S., Costa T., Cauda F (2019) The neural correlates of time: a meta-analysis of neuroimaging studies. *Journal of Cognitive Neuroscience* **31**:1796-1826
- Núñez R., Cooperrider K (2013) The tangle of space and time in human cognition. *Trends in Cognitive Sciences* **17**:220-229
- O'keefe J., Nadel L (1978) *The hippocampus as a cognitive map* Oxford University Press.
- Parkinson C., Liu S., Wheatley T (2014) A common cortical metric for spatial, temporal, and social distance. *Journal of Neuroscience* **34**:1979-1987

- Peer M., Salomon R., Goldberg I., Blanke O., Arzy S (2015) Brain system for mental orientation in space, time, and person. *Proceedings of the National Academy of Sciences* **112**:11072-11077
- Poppenk J., Evensmoen H. R., Moscovitch M., Nadel L (2013) Long-axis specialization of the human hippocampus. *Trends in Cognitive sciences* **17**:230-240
- Power J. D., Mitra A., Laumann T. O., Snyder A. Z., Schlaggar B. L., Petersen S. E (2014) Methods to detect, characterize, and remove motion artifact in resting state fMRI. *Neuroimage* **84**:320-341
- Protopapa F., Hayashi M. J., Kulashekhar S., van der Zwaag W., Battistella G., Murray M. M., ..., Buetti D. (2019) Chronotopic maps in human supplementary motor area. *PLOS Biology* **17**:e3000026
- Reznik D., Trampel R., Weiskopf N., Witter M. P., Doeller C. F (2023) Dissociating distinct cortical networks associated with subregions of the human medial temporal lobe using precision neuroimaging. *Neuron* **111**:2756-2772
- Richter M., Amunts K., Mohlberg H., Bludau S., Eickhoff S. B., Zilles K., Caspers S (2019) Cytoarchitectonic segregation of human posterior intraparietal and adjacent parieto-occipital sulcus and its relation to visuomotor and cognitive functions. *Cerebral Cortex* **29**:1305-1327
- Rizzo M., Vecera S. P (2002) Psychoanatomical substrates of Balint's syndrome. *Journal of Neurology, Neurosurgery & Psychiatry* **72**:162-178
- Robbe D (2023) Lost in time: Relocating the perception of duration outside the brain. *Neuroscience & Biobehavioral Reviews* **105312**
- Roberts W. A., Feeney M. C (2009) The comparative study of mental time travel. *Trends in Cognitive Sciences* **13**:271-277
- Robin J., Moscovitch M (2017) Details, gist and schema: hippocampal-neocortical interactions underlying recent and remote episodic and spatial memory. *Current Opinion in Behavioral Sciences* **17**:114-123
- Rolls E. T., Huang C. C., Lin C. P., Feng J., Joliot M (2020) Automated anatomical labelling atlas 3. *Neuroimage* **206**:116189
- Sestieri C., Shulman G. L., Corbetta M (2010) Attention to memory and the environment: functional specialization and dynamic competition in human posterior parietal cortex. *Journal of Neuroscience* **30**:8445-8456
- Sestieri C., Shulman G. L., Corbetta M (2017) The contribution of the human posterior parietal cortex to episodic memory. *Nature Reviews Neuroscience* **18**:183-192
- Shepard R. N., Judd S. A (1976) Perceptual illusion of rotation of three-dimensional objects. *Science* **191**:952-954
- Siegel A. W., White S. H (1975) The development of spatial representations of large-scale environments. *Advances in Child Development and Behavior* **10**:9-55
- Silani G., Lamm C., Ruff C. C., Singer T (2013) Right supramarginal gyrus is crucial to overcome emotional egocentricity bias in social judgments. *Journal of Neuroscience* **33**:15466-15476
- Sinha C., Gärdenfors P (2014) Time, space, and events in language and cognition: a comparative view. *Annals of the New York Academy of Sciences* **1326**:72-81
- Stark M (1996) Impairment of an egocentric map of locations: Implications for perception and action. *Cognitive Neuropsychology* **13**:481-524
- Stocker K (2012) The time machine in our mind. *Cognitive Science* **36**:385-420
- Stocker K. (2014) The theory of cognitive spacetime. *Metaphor and Symbol* **29**:71-93
- Strange B. A., Witter M. P., Lein E. S., Moser E. I (2014) Functional organization of the hippocampal longitudinal axis. *Nature Reviews Neuroscience* **15**:655-669
- Suddendorf T., Addis D. R., Corballis M. C (2009) Mental time travel and the shaping of the human mind. *Philosophical Transactions of the Royal Society B: Biological Sciences* **364**:1317-1324
- Summerfield C., Luyckx F., Sheahan H (2020) Structure learning and the posterior parietal cortex. *Progress in Neurobiology* **184**:101717

- Thavabalasingam S., O'Neill E. B., Lee A. C (2018) Multivoxel pattern similarity suggests the integration of temporal duration in hippocampal event sequence representations. *NeuroImage* **178**:136-146
- Theves S., Fernandez G., Doeller C. F (2019) The hippocampus encodes distances in multidimensional feature space. *Current Biology* **29**:1226-1
- Traugott E. C (1978) On the expression of spatio-temporal relations in language. *Universals of Human Language* **3**:369-400
- Tsao A., Yousefzadeh S. A., Meck W. H., Moser M. B., Moser E. I (2022) The neural bases for timing of durations. *Nature Reviews Neuroscience* **23**:646-665
- Tulving E (1984) Precis of elements of episodic memory. *Behavioral and Brain Sciences* **7**:223-238
- Tulving E. (2002) Episodic memory: From mind to brain. *Annual review of psychology* **53**:1-25
- Tversky B., Jamalian A (2021) Thinking tools: Gestures change thought about time. *Topics in Cognitive Science* **13**:750-776
- Viganò S., Bayramova R., Doeller C. F., Bottini R (2023) Mental search of concepts is supported by egocentric vector representations and restructured grid maps. *Nature Communications* **14**:8132
- Vogele K., Fink G. R (2003) Neural correlates of the first-person-perspective. *Trends in Cognitive Sciences* **7**:38-42
- Wagner A. D., Shannon B. J., Kahn I., Buckner R. L (2005) Parietal lobe contributions to episodic memory retrieval. *Trends in Cognitive Sciences* **9**:445-453
- Whitlock J. R (2017) Posterior parietal cortex. *Current Biology* **27**:R691-R695
- Winkler A. M., Ridgway G. R., Douaud G., Nichols T. E., Smith S. M (2016) Faster permutation inference in brain imaging. *Neuroimage* **141**:502-516
- Winkler A. M., Ridgway G. R., Webster M. A., Smith S. M., Nichols T. E (2014) Permutation inference for the general linear model. *Neuroimage* **92**:381-397
- Wittmann M (2013) The inner sense of time: how the brain creates a representation of duration. *Nature Reviews Neuroscience* **14**:217-223
- Woo C. W., Krishnan A., Wager T. D (2014) Cluster-extent based thresholding in fMRI analyses: pitfalls and recommendations. *Neuroimage* **91**:412-419
- Zacks J. M., Tversky B (2001) Event structure in perception and conception. *Psychological Bulletin* **127**:3
- Zipser D., Andersen R. A (1988) A back-propagation programmed network that simulates response properties of a subset of posterior parietal neurons. *Nature* **331**:679-684

Peer reviews

Reviewer #1 (Public review):

Summary:

In this fMRI study, the authors wished to assess neural mechanisms supporting flexible temporal construals. For this, human participants learned a story consisting of fifteen events. During fMRI, events were shown to them, and participants were instructed to consider the event from "an internal" or from "an external" perspective. The authors found distinct patterns of brain activity in the posterior parietal cortex (PPC) and anterior hippocampus for the internal and the external viewpoint. Specifically, activation in the posterior parietal cortex positively correlated with distance during the external-perspective task, but negatively during the internal-perspective task. The anterior hippocampus positively correlated with distance in both perspectives. The authors conclude that allocentric sequences are stored in the hippocampus, whereas egocentric sequences are supported by the parietal cortex.

Strengths:

The research topic is fascinating, and very few labs in the world are asking the question of how time is represented in the human brain. Working hypotheses have been recently formulated, and the work tackles them from the perspective of construals theory.

Weaknesses:

Although the work uses two distinct psychological tasks, the authors do not elaborate on the cognitive operationalization the tasks entail, nor the implication of the task design for the observed neural activation.

<https://doi.org/10.7554/eLife.107273.2.sa2>

Reviewer #2 (Public review):

Summary:

Xu et al. used fMRI to examine the neural correlates associated with retrieving temporal information from an external compared to internal perspective ('mental time watching' vs. 'mental time travel'). Participants first learned a fictional religious ritual composed of 15 sequential events of varying durations. They were then scanned while they either (1) judged whether a target event happened in the same part of the day as a reference event (external condition); or (2) imagined themselves carrying out the reference event and judged whether the target event occurred in the past or will occur in the future (internal condition). Behavioural data suggested that the perspective manipulation was successful: RT was positively correlated with sequential distance in the external perspective task, while a negative correlation was observed between RT and sequential distance for the internal perspective task. Neurally, the two tasks activated different regions, with the external task associated with greater activity in the supplementary motor area and supramarginal gyrus, and the internal condition with greater activity in default mode network regions. Of particular interest, only a cluster in the posterior parietal cortex demonstrated a significant interaction between perspective and sequential distance, with increased activity in this region for longer sequential distances in the external task but increased activity for shorter sequential distances in the internal task. Only a main effect of sequential distance was observed in the hippocampus head, with activity being positively correlated with sequential distance in both tasks. No regions exhibited a significant interaction between perspective and duration, although there was a main effect of duration in the hippocampus body with greater activity for longer durations, which appeared to be driven by the internal perspective condition. On the basis of these findings, the authors suggest that the hippocampus may represent event sequences allocentrically, whereas the posterior parietal cortex may process event sequences egocentrically.

Strengths:

The topic of egocentric vs. allocentric processing has been relatively under-investigated with respect to time, having traditionally been studied in the domain of space. As such, the current study is timely and has the potential to be important for our understanding of how time is represented in the brain in the service of memory. The study is well thought out and the behavioural paradigm is, in my opinion, a creative approach to tackling the authors' research question. A particular strength is the implementation of an imagination phase for the participants while learning the fictional religious ritual. This moves the paradigm beyond semantic/schema learning and is probably the best approach besides asking the participants to arduously enact and learn the different events with their exact timings in person. Importantly, the behavioural data point towards successful manipulation of internal vs.

external perspective in participants, which is critical for the interpretation of the fMRI data. The use of syllable length as a sanity check for RT analyses as well as neuroimaging analyses is also much appreciated.

Suggestions:

The authors have done a commendable job addressing my previous comments. In particular, the additional analyses elucidating the potential contribution of boundary effects to the behavioural data, the impact of incorporating RT into the fMRI GLMs, and the differential contributions of RT and sequential distance to neural activity (i.e., in PPC) are valuable and strengthen the authors' interpretation of their findings.

My one remaining suggestion pertains to the potential contribution of boundary effects. While the new analyses suggest that the RT findings are driven by sequential distance and duration independent of a boundary effect (i.e., Same vs. Different factor), I'm wondering whether the same applies to the neural findings? In other words, have the authors run a GLM in which the Same vs. Different factor is incorporated alongside distance and duration?

<https://doi.org/10.7554/eLife.107273.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this fMRI study, the authors wished to assess neural mechanisms supporting flexible "temporal construals". For this, human participants learned a story consisting of fifteen events. During fMRI, events were shown to them, and they were instructed to consider the event from "an internal" or from "an external" perspective. The authors found opposite patterns of brain activity in the posterior parietal cortex and the anterior hippocampus for the internal and the external viewpoint. They conclude that allocentric sequences are stored in the hippocampus, whereas egocentric sequences are used in the parietal cortex. The claims align with previous fMRI work addressing this question.

We appreciate the reviewer's concise summary of our research. We would like to offer two clarifications to prevent any potential misunderstandings.

First, the activity patterns in the parietal cortex and hippocampus are not entirely opposite across internal and external perspectives. Specifically, the activation level in the posterior parietal cortex shows a positive correlation with sequential distance during external-perspective tasks, but a negative correlation during internal-perspective tasks. In contrast, the activation level in the anterior hippocampus positively correlates with sequential distance, irrespective of the observer's perspective. Therefore, our results suggest that the parietal cortex, with its perspective-dependent activity, supports egocentric representation; the hippocampus, with its consistent activity across perspectives, supports allocentric representation.

Second, while some of our findings align with previous fMRI studies, to our knowledge, no prior research has explicitly investigated how the neural representation of time may vary depending on the observer's viewpoint. This gap in the literature is the primary motivation for our current study.

Strengths:

The research topic is fascinating, and very few labs in the world are asking the question of how time is represented in the human brain. Working hypotheses have been recently formulated, and this work seems to want to tackle some of them.

We appreciate the reviewer's acknowledgment of the theoretical significance of our study.

Weaknesses:

The current writing is fuzzy both conceptually and experimentally. I cannot provide a sufficiently well-informed assessment of the quality of the experimental work because there is a paucity of details provided in the report. Any future revisions will likely improve transparency.

(1) Improving writing and presentation:

The abstract and the introduction make use of loaded terms such as "construals", "mental timeline", "panoramic views" in very metaphoric and unexplained ways. The authors do not provide a comprehensive and scholarly overview of these terms, which results in verbiage and keywords/name-dropping without a clear general framework being presented. Some of these terms are not metaphors. They do refer to computational concepts that the authors should didactically explain to their readership. This is all the more important that some statements in the Introduction are misattributed or factually incorrect; some statements lack attributions (uncited published work). Once the theory, the question, and the working hypothesis are clarified, the authors should carefully explain the task.

We appreciate the reviewer's critics.

The formulation of the scientific question in the introduction is grounded in the spatial construals of time hypothesis and conceptual metaphor theory (e.g., Traugott, 1978; Lakoff & Johnson, 1980; see recent reviews by Núñez & Cooperrider, 2013; Bender & Beller, 2014). These frameworks were originally developed through analyses of how spatial metaphors are used to describe temporal concepts in natural language. Consequently, it is theoretically motivated and largely unavoidable to introduce the two primary temporal construals—mental time travel and mental time watching—using metaphorical expressions.

However, we do agree with the reviewer that the introduction in the original manuscript was overly long and that the working hypothesis was not clearly stated. In the revised manuscript, we have streamlined the introduction and substantially revised the following two paragraphs to clarify the formulation of our working hypothesis (Pages 5-6):

“Recent studies have already begun to investigate the neural representation of the memorized event sequence (e.g., Deuker et al., 2016; Thavabalasingam et al., 2018; Bellmund et al., 2019, 2022; see reviews by Cohn-Sheehy & Ranganath, 2017; Bellmund et al., 2020). Yet, the neural mechanisms that enable the brain to construct distinct construals of an event sequence remain largely unknown. Valuable insights may be drawn from research in the spatial domain, which differentiates the neural representation in allocentric and egocentric reference frames. According to an influential neurocomputational model (Byrne et al., 2007; Bicanski & Burgess, 2018; Bicanski & Burgess, 2020), allocentric and egocentric spatial representations are dissociable in the brain—they are respectively implemented in the medial temporal lobe (MTL)—including the hippocampus—and the parietal cortex. Various egocentric representations in the parietal cortex derived from different viewpoints can be transformed and integrated into a unified allocentric representation and stored in the MTL (i.e., bottom-up process). Conversely, the allocentric representation in the MTL can serve as a template for reconstructing diverse egocentric representations across different viewpoints in the parietal cortex (i.e., top-down process).”

“In line with the spatial construals of time hypothesis, several authors have recently suggested that such mutually engaged egocentric and allocentric reference frames (in the parietal cortex and the medial temporal lobe, respectively) proposed in the spatial domain might also apply to the temporal one (e.g., Gauthier & van Wassenhove, 2016ab; Gauthier et al., 2019, 2020; Bottini & Doeller, 2020). If this hypothesis holds, it could explain how the brain flexibly generates diverse construals of the same event sequence. Specifically, the hippocampus may encode a consistent representation of an event sequence that is independent of whether an individual adopts an internal or external perspective, reflecting an allocentric representation of time. In contrast, parietal cortical representations are expected to vary flexibly with the adopted perspective that is shaped by task demands, reflecting an egocentric representation of time.”

In the revised manuscript, we also corrected statements in the Introduction that may have been misattributed (see Reviewer 2, comment 4(ii)) and added several relevant and important publications.

(2) The experimental approach lacks sufficient details to be comprehensible to a general audience. In my opinion, the results are thus currently uninterpretable. I highlight only a couple of specific points (out of many). I recommend revision and clarification.

(a) No explanation of the narrative is being provided. The authors report a distribution of durations with no clear description of the actual sequence of events. The authors should provide the text that was used, how they controlled for low-level and high-level linguistic confounds.

We thank the reviewer for the suggestions. The event sequence for the odd-numbered participants is shown in the original Figure 1. In the revised manuscript, we added to Figure 1 the figure supplement 1 to illustrate the actual sequence of events for the participants with both odd and even numbers. We also added the narratives used in the reading phase of the learning procedures for the participants with both odd and even numbers (Figure 1—source data 1).

To control for low-level linguistic confounds, we included the number of syllables as a covariate in the first-level general linear model in the fMRI analysis. To address high-level linguistic confounds, such as semantic information (which is difficult to quantify), we randomly assigned event labels to the 15 events twice, creating two counterbalanced versions for participants with even and odd numbers (see Comment 2b below).

(b) The authors state, "we randomly assigned 15 phrases to the events twice". It is impossible to comprehend what this means. Were these considered stimuli? Controls? IT is also not clear which event or stimulus is part of the "learning set" and whether these were indicated to be such to participants.

We apologize for any confusion in the Results section and the legend of Figure 1. Our motivation was explained in the "Stimuli" section of the Methods. In the revised manuscript, we have clarified this by adding an explanation to the legend of Figure 1 and including the supplementary Figure 1: "To minimize potential confounds between the semantic content of the event phrases and the temporal structure of the events, we randomly assigned the phrases to the events, creating two versions for participants with even and odd ID numbers. Both versions can be seen in Figure1—figure supplement 1 and Figure 1—source data 1."

(c) The left/right counterbalancing is not being clearly explained. The authors state that there is counterbalancing, but do not sufficiently explain what it means concretely in the experiment. If a weak correlation exists between sequential position and distance, it also means that the position and the distance have not been equated within. How do the authors control for these?

We thank the reviewer for highlighting this point and apologize for the lack of clarity in the original manuscript. In the current version (Page 40), we have provided further clarification: “We carefully selected two sets of 20 event pairs from the 210 possible combinations, assigning them to the odd and even runs of the fMRI experiment. Using a brute-force search, we identified 20 pairs in which sequential distance showed only weak correlations with positional information for both reference and target events (ranging from 1 to 15), as well as with behavioral responses (Same vs. Different or Future vs. Past, coded as 0 and 1), with all correlation coefficients below 0.2. At the same time, we balanced the proportion of correct responses across conditions: for the external-perspective task, Same/Different = 11/9 and 12/8; for the internal-perspective task, Future/Past = 12/8 and 8/12. Under these constraints, the sequential distances in both sets ranged from 1 to 5. To further mitigate spatial response biases, we pseudorandomized the left/right on-screen positions of the two response options within each task block, while ensuring an equal number of correct responses mapped to the left and right buttons (i.e., 10 per block).”

The event pairs we selected already represent the best possible choice given all the criteria we aimed to satisfy. It is impossible to completely eliminate all potential correlations. For instance, if the target event occurs near the beginning of the day, it will tend to fall in the past, whereas if it occurs near the end of the day, it is more likely to fall in the future. To further ensure that the significant results were not driven by these weak confounding factors, we constructed another GLM that included three additional parametric modulators: the sequence position of the target event (ranging from 1 to 15) and the behavioral responses (Future vs. Past in the internal-perspective task; Same vs. Different in the external-perspective task, coded as 0 and 1). The significant findings were unaffected.

(d) The authors used two tasks. In the "external perspective" one, the authors asked participants to report whether events were part of the same or a different part of the day. In the "internal perspective one", the authors asked participants to project themselves to the reference event and to determine whether the target event occurred before or after the projected viewpoint. The first task is a same/different recognition task. The second task is a temporal order task (e.g., Arzy et al. 2009). These two tasks are radically different and do not require the same operationalization. The authors should minimally provide a comprehensive comparison of task requirements, their operationalization, and, more importantly, assess the behavioral biases inherent to each of these tasks that may confound brain activity observed with fMRI.

We understand the reviewer's concern. We agree that there is a substantial difference between the two tasks. However, the primary goal of this study was not to directly compare these tasks to isolate a specific cognitive component. Rather, the neural correlates of temporal distance were first identified as brain regions showing a significant correlation between neural activity and temporal distance using the parametric modulation analysis. We then compared these neural correlates between the two tasks. Therefore, any general differences between the tasks should not be a confound for our main results. Our aim was to examine whether the hippocampal representation of temporal distance remains consistent across different perspectives, and whether the parietal representation of temporal distance varies as a function of the perspective adopted.

Therefore, the main aim of our task manipulation was to ensure that participants adopted either an external or an internal perspective on the event sequence, depending on the task condition. In the Introduction (Pages 6–7), we clarify this manipulation as follows: “In the externalperspective task, participants localized events with respect to external temporal boundaries, judging whether the target event occurred in the same or a different part of the day as the reference event. In the internal-perspective task, participants were instructed to mentally project themselves into the reference event and localize the target event relative to

their own temporal point, judging whether the target event happened in the future or the past of the reference event (see Methods for details of the scanning procedure).”

We believe this task manipulation was successful. Behaviorally, the two tasks showed opposite correlations between reaction time and temporal distance, resembling the symbolic distance versus mental scanning effect. Neurally, contrasting the internal- and external-perspective tasks revealed activation of the default mode network, which is known to play a central role in self-projection (Buckner et al., 2017).

(e) The authors systematically report interpreted results, not factual data. For instance, while not showing the results on behavioral outcomes, the authors directly interpret them as symbolic distance effects.

Thank you for this comment. In the original paper, we reported the relevant statistics before our interpretation: “Sequential Distance was correlated positively with RT in the external-perspective task ($z = 3.80$, $p < 0.001$) but negatively in the internal-perspective task ($z = -3.71$, $p < 0.001$).” However, they may have been difficult to notice, and we are including a figure for the RT analysis in the revised manuscript.

Crucially, the authors do not comment on the obvious differences in task difficulty in these two tasks, which demonstrates a substantial lack of control in the experimental design. The same/different task (task 1 called “external perspective”) comes with known biases in psychophysics that are not present in the temporal order task (task 2 called “internal perspective”). The authors also did not discuss or try to match the performance level in these two tasks. Accordingly, the authors claim that participants had greater accuracy in the external (same/different) task than in the internal task, although no data are shown and provided to support this report. Further, the behavioral effect is trivialized by the report of a performance accuracy trade off that further illustrates that there is a difference in the task requirements, preventing accurate comparison of the two tasks.

As noted in Question 2d, we acknowledge the substantial difference between the two tasks. However, the primary goal of this study was not to directly compare these tasks to isolate a specific cognitive component. Instead, we first identified the neural correlates of temporal distance as brain regions showing a significant correlation between neural activity and temporal distance, independent of task demands. We then compared these neural correlates across the two task conditions, which were designed to engage different temporal perspectives. Therefore, any general differences between the tasks should not be a confound for our main findings and interpretation.

Our aim was to investigate whether the hippocampal representation of temporal distance remains consistent across different perspectives and whether the parietal representation of temporal distance varies as a function of the perspective adopted. We do not see how this doubledissociation pattern could be explained by differences in task difficulty.

While we do not consider the overall difference in task difficulty between the two tasks to be a confounding factor, we acknowledge the potential confound posed by variations in task difficulty across temporal distances (1 to 5). This concern arises from the similarity between the activity patterns in the posterior parietal cortex and reaction time across temporal distances. To address this, we conducted control analyses to test this hypothesis (see the second and third points from Reviewer 2 for details).

On page 8, we present the behavioral accuracy data: “Participants showed significantly higher accuracy in the external-perspective task than in the internal-perspective task (external-perspective task: $M = 93.5\%$, $SD = 4.7\%$; internal-perspective task: $M = 89.5\%$, $SD = 8.1\%$; paired $t(31) = 3.33$, $p = 0.002$).”

All fMRI contrasts are also confounded by this experimental shortcoming, seeing as they are all reported at the interaction level across a task. For instance, in Figure 4, the authors report a significant beta difference between internal and external tasks. It is impossible to disentangle whether this effect is simply due to task difference or to an actual processing of the duration that differs across tasks, or to the nature of the representation (the most difficult to tackle, and the one chosen by the authors).

We thank the reviewer for pointing out this important issue. Like temporal distance, the neural correlates of duration were not derived from a direct contrast between the two tasks. Instead, they were identified by detecting brain regions showing a significant correlation between neural activity and the implied duration of each event using the parametric modulation analysis. Therefore, what is shown in Figure 4 reflects the significant differences in these neural correlations with duration between the two tasks.

The observed difference in the neural representation of duration between the two tasks was unexpected. In the original manuscript, we provided a post hoc explanation: “Since the externalperspective task in the current study encouraged the participants to compare the event sequence with the external parallel temporal landmarks, duration representation in the hippocampus may be dampened.”

However, we agree that this difference might also arise from other factors distinguishing the two tasks. In the revised manuscript, we have clarified this possibility as follows: “The difference in duration representation between the two tasks remains open to interpretation. One possible explanation is that the hippocampus is preferentially involved in memory for durations embedded within event sequences (see review by Lee et al., 2020). In the internal-perspective task, participants indeed localized events within the event sequence itself. In contrast, the externalperspective task encouraged participants to compare the event sequence with external temporal landmarks, which may have attenuated the hippocampal representation of duration.”

Conclusion:

In conclusion, the current experimental work is confounded and lacks controls. Any behavioral or fMRI contrasts between the two proposed tasks can be parsimoniously accounted for by difficulty or attentional differences, not the claim of representational differences being argued for here.

We hope that our explanations and clarifications above adequately address the reviewer’s concerns. We would like to reiterate that we did not directly compare the two tasks. Rather, we first identified the neural representations of sequential distance and duration, and then examined how these representations differed across tasks. It is unclear to us how the overall difference in task difficulty or attentional demands could lead to the observed pattern of results.

By determining where the neural representations were consistent and where they diverged, we were able to differentiate brain regions that encode temporal information allocentrically from those that represent temporal information in a perspective-dependent manner, modulated by task demands.

Reviewer #2 (Public review):

Summary:

Xu et al. used fMRI to examine the neural correlates associated with retrieving temporal information from an external compared to internal perspective ('mental time watching' vs. 'mental time travel'). Participants first learned a fictional religious ritual composed of

15 sequential events of varying durations. They were then scanned while they either (1) judged whether a target event happened in the same part of the day as a reference event (external condition); or (2) imagined themselves carrying out the reference event and judged whether the target event occurred in the past or will occur in the future (internal condition). Behavioural data suggested that the perspective manipulation was successful: RT was positively correlated with sequential distance in the external perspective task, while a negative correlation was observed between RT and sequential distance for the internal perspective task. Neurally, the two tasks activated different regions, with the external task associated with greater activity in the supplementary motor area and supramarginal gyrus, and the internal condition with greater activity in default mode network regions. Of particular interest, only a cluster in the posterior parietal cortex demonstrated a significant interaction between perspective and sequential distance, with increased activity in this region for longer sequential distances in the external task, but increased activity for shorter sequential distances in the internal task. Only a main effect of sequential distance was observed in the hippocampus head, with activity being positively correlated with sequential distance in both tasks. No regions exhibited a significant interaction between perspective and duration, although there was a main effect of duration in the hippocampus body with greater activity for longer durations, which appeared to be driven by the internal perspective condition. On the basis of these findings, the authors suggest that the hippocampus may represent event sequences allocentrically, whereas the posterior parietal cortex may process event sequences egocentrically.

We sincerely appreciate the reviewers for providing an accurate, comprehensive, and objective summary of our study.

Strengths:

The topic of egocentric vs. allocentric processing has been relatively under-investigated with respect to time, having traditionally been studied in the domain of space. As such, the current study is timely and has the potential to be important for our understanding of how time is represented in the brain in the service of memory. The study is well thought out, and the behavioural paradigm is, in my opinion, a creative approach to tackling the authors' research question. A particular strength is the implementation of an imagination phase for the participants while learning the fictional religious ritual. This moves the paradigm beyond semantic/schema learning and is probably the best approach besides asking the participants to arduously enact and learn the different events with their exact timings in person. Importantly, the behavioural data point towards successful manipulation of internal vs. external perspective in participants, which is critical for the interpretation of the fMRI data. The use of syllable length as a sanity check for RT analyses, as well as neuroimaging analyses, is also much appreciated.

We thank the reviewer for the positive and encouraging comments.

Weaknesses/Suggestions:

Although the design and analysis choices are generally solid, there are a few finer details/nuances that merit further clarification or consideration in order to strengthen the readers' confidence in the authors' interpretation of their data.

(1) Given the known behavioural and neural effects of boundaries in sequence memory, I was wondering whether the number of traversed context boundaries (i.e., between morning-afternoon, and afternoon-evening) was controlled for across sequential length in the internal perspective condition? Or, was it the case that reference-target event pairs with higher sequential numbers were more likely to span across two parts of the day

compared to lower sequential numbers? Similarly, did the authors examine any potential differences, whether behaviourally or neurally, for day part same vs. day part different external task trials?

We thank the reviewer for the thoughtful comments. When we designed the experiment, we minimized the correlation between the sequential distance between the target and reference events and whether the reference and target events occurred within the same or different parts of the day (coded as Same = 0, Different = 1). The point-biserial correlation coefficient between these two variables across all the trials within the same run were controlled below 0.2.

To investigate the effect of day-part boundaries on behavior, as well as the contribution of other factors, we conducted a new linear mixed-effects model analysis incorporating four additional variables. They are whether the target and the reference events are within the same or different parts of the day (i.e., Same vs. Different), whether the target event is in the future or the past of the reference event (i.e., Future vs. Past), and the interactions of the two factors with Task Type (i.e., internal- vs. external-perspective task).

The results are largely the same as the original one in the table: There was a significant main effect of Syllable Length, and the interaction effects between Task Type and Sequence Distance and between Task Type and Duration remain significant. What's new is we also found a significant interaction effect between Task Type and Same vs. Different.

As shown in the Figure 2—figure supplement 1, this Same vs. Different effect was in line with the effect of Sequential Distance, with two events in the same and different parts of the day corresponding to the short and long sequential distances. Given that Sequential Distance had already been considered in the model, the effect of parts of the day should result from the boundary effect across day parts or the chunking effect within day parts, i.e., the sequential distance across different parts of the day was perceived longer while the sequential distance within the same parts of the day was perceived shorter. We have incorporated these findings into the manuscript.

Neurally, to further verify that the significant effects of sequential distance were not driven by its weak correlation with the Same/Different judgment or other potential confounding factors, we constructed another GLM that incorporated three additional parametric modulators: the sequence position of the target event (ranging from 1 to 15) and the behavioral responses (Future vs. Past in the internal-perspective task; Same vs. Different in the external-perspective task, coded as 0 and 1). The significant findings were unaffected.

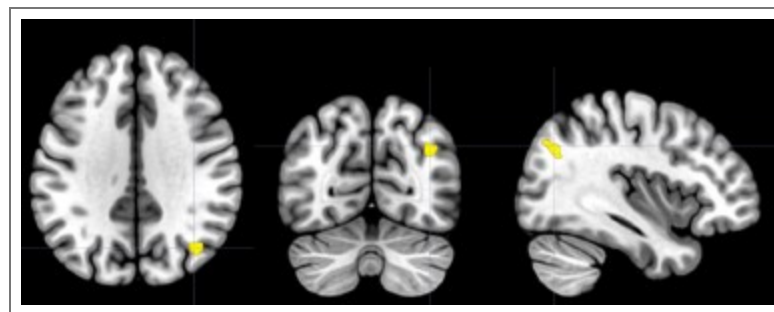
(2) I would appreciate further insight into the authors' decision to model their task trials as stick functions with duration 0 in their GLMs, as opposed to boxcar functions with varying durations, given the potential benefits of the latter (e.g., Grinband et al., 2008). I concur that in certain paradigms, RT is considered a potential confound and is taken into account as a nuisance covariate (as the authors have done here). However, given that RTs appear to be critical to the authors' interpretation of participant behavioural performance, it would imply that variations in RT actually reflect variations in cognitive processes of interest, and hence, it may be worth modelling trials as boxcar functions with varying durations.

We appreciate the reviewer's insightful comment on this important issue. Whether to control for RT's influence on fMRI activation is indeed a long-standing paradox. On the one hand, RT reflects underlying cognitive processes and therefore should not be fully controlled for. On the other hand, RT can independently influence neural activity, as several brain networks vary with RT irrespective of the specific cognitive process involved—a domain-general effect. For example, regions within the multiple-demand network are often positively correlated with RT across different cognitive domains.

Our strategy in the manuscript is to first present the results without including RT as a control variable and then examine whether the effects are preserved after controlling for RT. In the revised manuscript, we have clarified this approach (Page 13): “Here, changes in activity levels within the PPC were found to align with RT. Whether to control for RT’s influence on fMRI activation represents a well-known paradox. On the one hand, RT reflects underlying cognitive processes and therefore should not be fully controlled for. On the other hand, RT can independently influence neural activity, as several brain networks vary with RT irrespective of the specific cognitive process involved—a domain-general effect. For instance, regions within the multiple-demand network are often positively correlated with RT and task difficulty across diverse cognitive domains (e.g., Fedorenko et al., 2013; Mumford et al., 2024). To evaluate the second possibility, we conducted an additional control analysis by including trial-by-trial RT as a parametric modulator in the first-level model (see Methods). Notably, the same PPC region remained the only area in the entire brain showing a significant interaction between Task Type and Sequential Distance (voxel-level $p < 0.001$, clusterlevel FWE-corrected $p < 0.05$). This finding indicates that PPC activity cannot be fully attributed to RT. Furthermore, we do not interpret the effect as reflecting a domain-general RT influence, as regions within the multiple-demand system—typically sensitive to RT and task difficulty—did not exhibit significant activation in our data.”

The reason we did not use boxcar functions with varying durations in our original manuscript is that we also applied parametric modulation in the same model. In the parametric modulation, all parametric modulators inherit the onsets and durations of the events being modulated. Consequently, the modulators would also take the form of boxcar functions rather than stick functions—the height of each boxcar reflecting the parameter value and its length reflecting the RT. We were uncertain whether this approach would be appropriate, as we have not encountered other studies implementing parametric modulation in this manner.

For exploratory purposes, we also conducted a first-level analysis using boxcar functions with variable durations. The same PPC region remained the strongest area in the entire brain that shows an interaction effect between Task Type and Sequential Distance. However, the cluster size was slightly reduced (voxel-level $p < 0.001$, cluster-level FWE-corrected $p = 0.0610$; see the Author response image 1 below). The cross indicates the MNI coordinates at [38, -69, 35], identical to those shown in the main results (Figure 4A).

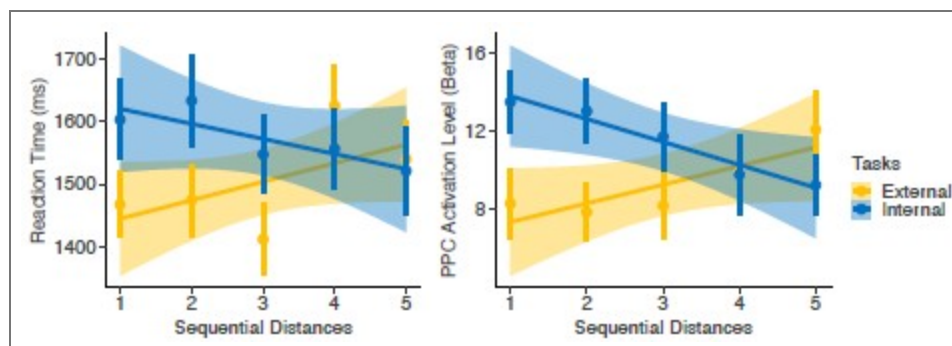


Author response image 1.

(3) *The activity pattern across tasks and sequential distance in the posterior parietal cortex appears to parallel the RT data. Have the authors examined potential relationships between the two (e.g., individual participant slopes for RT across sequential distance vs. activity betas in the posterior parietal cortex)?*

We thank the reviewer for this helpful suggestion. As shown in the Author response image 2, the interaction between Task Type and Sequential Distance was a stronger predictor of PPC

activation than of RT. Because PPC activation and RT are measured on different scales, we compared their standardized slopes (standardized β) measuring the change in a dependent variable in terms of standard deviations for a one-standard-deviation increase in an independent variable. The standardized β for the Task Type $\times$ Sequential Distance interaction was -0.30 (95% CI $[-0.42, -0.19]$) for PPC activation and -0.21 (95% CI $[-0.30, -0.13]$) for RT. The larger standardized effect for PPC activation indicates that the Task Type $\times$ Sequential Distance interaction was a stronger predictor of neural activation than of behavioral RT.



Author response image 2.

A more relevant question is whether PPC activation can be explained by temporal information (i.e., the sequential distance) independently of RT. To test this, we included both Sequential Distance and RT in the same linear mixed-effects model predicting PPC Activation Level. As shown in the Author response table 1, although RT independently influenced PPC activation ($F(1, 288) = 4.687, p = 0.031$), the interaction between Task Type and Sequential Distance was a much stronger independent predictor ($F(1, 290) = 19.319, p < 0.001$).

Fixed Effects ⁴	df	F	p
RT	1, 288	4.687	0.031
Sequential Distance	1, 283	0.420	0.518
Task Type	1, 284	9.853	0.002
Task Type $\times$ RT	1, 284	0.385	0.535
Task Type $\times$ Sequential Distance	1, 290	19.319	< 0.001

Author response table 1. PPC Activation Level Predicted by Sequential Distance and RT

(3) Linear Mixed Model Formula: $\text{PPC Activation Level} \sim 1 + \text{Task Type} * (\text{Sequential Distance} + \text{RT}) + (1 \mid \text{Participant})$

(4) There were a few places in the manuscript where the writing/discussion of the wider literature could perhaps be tightened or expanded. For instance:

(i) On page 16, the authors state 'The negative correlation between the activation level in the right PPC and sequential distance has already been observed in a previous fMRI study (Gauthier & van Wassenhove, 2016b). The authors found a similar region (the reported MNI coordinate of the peak voxel was 42, -70, 40, and the MNI coordinate of the peak voxel in the present study was 39, -70, 35), of which the activation level went up

when the target event got closer to the self-positioned event. This finding aligns with the evidence suggesting that the posterior parietal cortex implements egocentric representations.' Without providing a little more detail here about the Gauthier & van Wassenhove study and what participants were required to do (i.e., mentally position themselves at a temporal location and make 'occurred before' vs. 'occurred after' judgements of a target event), it could be a little tricky for readers to follow why this convergence in finding supports a role for the posterior parietal cortex in egocentric representations.

We appreciate the reviewer's comments. In the revised manuscript, we have provided a more detailed explanation of Gauthier and van Wassenhove's study (Page 17): "The negative correlation between the activation level in the right PPC and sequential distance has already been observed in a previous fMRI study by Gauthier & van Wassenhove (2016b). In their study, the participants were instructed to mentally position themselves at a specific time point and judge whether a target event occurred before or after that time point. The authors identified a similar brain region (reported MNI coordinates of the peak voxel: 42, -70, 40), closely matching the activation observed in the present study (MNI coordinates of the peak voxel: 39, -70, 35). In both studies, activation in this region increased as the target event approached the self-positioned time point, which aligns with the evidence suggesting that the posterior parietal cortex implements egocentric representations."

(ii) Although the authors discuss the Lee et al. (2020) review and related studies with respect to retrospective memory, it is critical to note that this work has also often used prospective paradigms, pointing towards sequential processing being the critical determinant of hippocampal involvement, rather than the distinction between retrospective vs. prospective processing.

We sincerely thank the reviewer for highlighting these important points. In response, we have revised the section of the Introduction discussing the neural underpinnings of duration (Pages 3-4). "Neurocognitive evidence suggests that the neural representation of duration engages distinct brain systems. The motor system—particularly the supplementary motor area—has been associated with prospective timing (e.g., Protopapa et al., 2019; Nani et al., 2019; De Kock et al., 2021; Robbe, 2023), whereas the hippocampus is considered to support the representation of duration embedded within an event sequence (e.g., Barnett et al., 2014; Thavabalasingam et al., 2018; see also the comprehensive review by Lee et al., 2020)."

(iii) The authors make an interesting suggestion with respect to hippocampal longitudinal differences in the representation of event sequences, and may wish to relate this to Montagrin et al. (2024), who make an argument for the representation of distant goals in the anterior hippocampus and immediate goals in the posterior hippocampus.

We thank the reviewer for bringing this intriguing and relevant study to our attention. In the Discussion of the manuscript, we have incorporated it into our discussion (Page 21): "Evidence from the spatial domain has suggested that the anterior hippocampus (or the ventral rodent hippocampus) implements global and gist-like representations (e.g., larger receptive fields), whereas the posterior hippocampus (or the dorsal rodent hippocampus) implements local and detailed ones (e.g., finer receptive fields) (e.g., Jung et al., 1994; Kjelstrup et al., 2008; Collin et al., 2015; see reviews by Poppenk et al., 2013; Robin & Moscovitch, 2017; see Strange et al., 2014 for a different opinion). Recent evidence further shows that the organizational principle observed along the hippocampal long axis may also extend to the temporal domain (Montagrin et al., 2024). In that study, the anterior hippocampus showed greater activation for remote goals, whereas the posterior hippocampus was more strongly engaged for current goals, which are presumed to be represented in finer detail."

Reviewing Editor Comments:

While both reviewers acknowledged the significance of the topic, they raised several important concerns. We believe that providing conceptual clarification, adding important methodological details, as well as addressing potential confounds will further strengthen this paper.

We thank the editor for the suggestions.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Please, provide the actual ethical approval #.

We have added the ethical approval number in the revised manuscript (P 36): “The ethical committee of the University of Trento approved the experimental protocol (Approval Number 2019-018),”

(2) Thirty-two participants were tested. Please report how you estimated the sample size was sufficient to test your working hypothesis.

We thank the editor for pointing out this omission. In the revised manuscript, we have added an explanation for our choice of sample size (p. 36): “The sample size was chosen to align with the upper range of participant numbers reported in previous fMRI studies that successfully detected sequence or distance effects in the hippocampus (N = 15–34; e.g., Morgan et al., 2011; Howard et al., 2014; Deuker et al., 2016; Garvert et al., 2017; Theves et al., 2019; Park et al., 2021; Cristoforetti et al., 2022).”

(3) All MRI figures: please orient the reader; left/right should be stated.

In the revised manuscript, we have added labels to all MRI figures to indicate the left and right hemispheres.

(4) In Figure 3A-B, the clear lateralization of the activation is not discussed in the Results or in the Discussion. Was it predicted?

We thank the editors for highlighting this important point regarding hemispheric lateralization. The right-lateralization observed in our findings is indeed consistent with previous literature. In the revised manuscript, we have expanded our discussion to emphasize this aspect more clearly.

For the parietal cortex, we now note (Page 17-18): “The negative correlation between activation in the right posterior parietal cortex (PPC) and sequential distance has previously been reported in an fMRI study by Gauthier and van Wassenhove (2016b). In their paradigm, participants were instructed to mentally position themselves at a specific time point and judge whether a target event occurred before or after that point. The authors identified a similar region (peak voxel MNI coordinates: 42, -70, 40), closely corresponding to the activation observed in the present study (peak voxel MNI coordinates: 39, -70, 35). In both studies, activation in this region increased as the target event approached the self-positioned time point, consistent with evidence suggesting that the posterior parietal cortex supports egocentric representations. Neuropsychological studies have further shown that patients with lesions in the bilateral or right PPC exhibit ‘egocentric disorientation’ (Aguirre & D’Esposito, 1999), characterized by an inability to localize objects relative to themselves (e.g., Case 2: Levine et al., 1985; Patient DW: Stark, 1996; Patients MU: Wilson et al., 1997, 2005).”

For the hippocampus, we have added (Page 19): “Previous research has shown that hippocampal activation correlates with distance (e.g., Morgan et al., 2011; Howard et al., 2014; Garvert et al., 2017; Theves et al., 2019; Viganò et al., 2023), and that distributed hippocampal

activity encodes distance information (e.g., Deuker et al., 2016; Park et al., 2021). Most studies have reported hippocampal ePects either bilaterally or predominantly in the right hemisphere, whereas only one study (Morgan et al., 2011) found the ePect localized to the left hippocampus.”

<https://doi.org/10.7554/eLife.107273.2.sa0>